

**IS 9473 : 2002**

(Reaffirmed 2014)

*भारतीय मानक*

श्वसन संरक्षी युक्तियाँ — कण सुरक्षा के लिए  
फिल्टरिंग हॉफ मास्क — विशिष्टि  
( पहला पुनरीक्षण )

*Indian Standard*

RESPIRATORY PROTECTIVE DEVICES —  
FILTERING HALF MASKS TO PROTECT AGAINST  
PARTICLES — SPECIFICATION

( *First Revision* )

ICS 13.340.30

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**BUREAU OF INDIAN STANDARDS**  
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NEW DELHI 110002

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Price Group 11

## Industrial Safety and Chemical Hazards Sectional Committee, CHD 8

### FOREWORD

This Indian Standard (First Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Industrial Safety and Chemical Hazards Sectional Committee had been approved by the Chemical Division Council.

Filtering half masks are intended to provide adequate sealing on the face of the wearer of a respiratory protective device against the ambient atmosphere, when the skin is dry or moist and when the head is moved. Air enters the filtering half mask and passes directly to the nose and mouth area of the facepiece or, via an inhalation valve, if fitted. The exhaled air flows through the filter material and/or an exhalation valve (if fitted) directly to the ambient atmosphere.

These devices can be designed to protect against solid and water-based aerosols only or against both solid and liquid aerosols where a solid aerosol is defined as a suspension of solid particles in air, a liquid aerosol is defined as a suspension of liquid droplets in air and a water based aerosol is defined as one which is produced from solutions and/or suspensions of solid materials in water such that the only hazardous component is attributable to the solid material.

This standard was originally published in 1980 based on the standard "Title 30 — Mineral, Resources" issued by U. S. Department of Health, Education and Welfare, Public Health Services, National Institute for Occupational Safety and Health (NIOSH). With a view to update the standard due to the technological changes that took place in the last two decades, the Committee felt a need to revise the standard and to align it with EN 149 : 1991 'Respiratory protective devices — Filtering half masks to protect against particles — Requirements, testing, marking'. There is no ISO standard on this subject.

The composition of the Committee responsible for formulating this standard is given at Annex B.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value shall be the same as that of the specified value in this standard.

*Indian Standard*  
**RESPIRATORY PROTECTIVE DEVICES —  
FILTERING HALF MASKS TO PROTECT AGAINST  
PARTICLES — SPECIFICATION**  
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## 1 SCOPE

1.1 This standard prescribes requirements and method of sampling and tests for filtering half masks as respiratory protective devices against particles except for escape purposes.

NOTE — Laboratory and practical performance tests are included for the assessment of compliance with the requirements.

## 2 REFERENCES

The Indian Standards listed below contain provisions, which through reference in this text constitute provisions of this Indian Standard. At the time of publication, the editions indicated were valid. All standards are subject to revisions, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated below:

| <i>IS No.</i> | <i>Title</i>                                                                  |
|---------------|-------------------------------------------------------------------------------|
| 4905 : 1968   | Methods of random sampling                                                    |
| 8347 : 1977   | Glossary of terms relating to respiratory protective devices                  |
| 14166 : 1994  | Respiratory protective devices — Full face masks — Specification              |
| 14746 : 1999  | Respiratory protective devices — Half masks and quarter masks — Specification |

## 3 TERMINOLOGY

3.1 For the purpose of this standard, the definitions given in IS 8347 and the term given below shall apply.

### 3.1.1 *Filtering Half Mask*

A filtering half mask is one which covers the nose and mouth and possibly the chin and

- a) consists entirely or substantially of filter material, or
- b) comprises a face piece in which the main filter(s) form an inseparable part of the device and where the pre-filter may be replaceable.

## 4 CLASS

4.1 Filtering half masks shall be of following three classes:

- a) FFP1,

- b) FFP2, and

- c) FFP3.

NOTE — FFP1 filters are intended for use against solid particles only. FFP2 and FFP3 filters are sub-divided according to their ability to remove both solid and liquid particles or solid particles only. The protection provided by FFP2 filter or FFP3 filter includes that provided by the corresponding filter of lower class or classes.

## 5 REQUIREMENTS

### 5.1 Material

5.1.1 When subjected to the conditioning described in A-1, none of the filtering half masks shall have suffered deformation of the face piece or straps. Three filtering half masks shall be tested.

5.1.2 When subjected to conditioning in accordance with A-1.1 and A-1.2, the filtering half mask shall not collapse.

5.1.3 Any material from the filter media released by the air flow through the filter should not constitute a hazard or nuisance for the wearer.

### 5.1.4 *Metal*

The use of aluminium, magnesium and titanium or alloys containing such proportions of these metals as will, on impact, give rise to frictional sparks capable of igniting flammable gas mixtures for exposed parts (those which may be subjected to impact during the use of apparatus) shall be restricted to a minimum.

### 5.2 Cleaning and Disinfecting

The materials used shall withstand the cleaning and disinfecting agents recommended by the manufacturer.

NOTE — This is applicable if the filtering half mask is designed for more than a single shift (not designed for single use only).

### 5.3 Practical Performance Test

The filtering half mask shall undergo practical performance tests under realistic conditions. These general tests serve the purpose of checking the equipment for imperfections that cannot be determined by the tests described elsewhere in this standard. Where a filtering half mask is to be used for filtering devices, testing shall be in accordance with A-2.

When practical performance tests show the apparatus has imperfections related to wearer's acceptance, full

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details of those parts of practical performance tests, which revealed these imperfections, shall be provided. This will enable during subsequent testing of the masks to repeat the tests and assess the results thereof.

### 5.4 Leakage

#### 5.4.1 Total Inward Leakage

5.4.1.1 When tested according to A-3, the filtering half masks fitted in accordance with the manufacturer's instruction, at least 46 out of 50 individual exercise results (that is, 10 subjects  $\times$  5 exercises) for the total inward leakage:

- shall not be greater than 25 percent for FFP1, 11 percent for FFP2 and 5 percent for FFP3; and
- at least 8 out of 10 individual wearer arithmetic means for the total inward leakage shall not be greater than 22 percent for FFP1, 8 percent for FFP2 and 2 percent for FFP3.

#### NOTES

1 The total inward leakage consists of three components: (a) face seal leakage, (b) exhalation valve leakage (if exhalation valve fitted), and (c) filter penetration.

2 The test indicates that the filtering half mask can be used by the wearer to protect with high probability against the potential hazard to be expected.

#### 5.4.2 Penetration of Filter Material

The penetration of the filter of the filtering half mask shall meet the requirements given in Table 1.

**Table 1 Penetration of Filter Material**

| Sl No. | Characteristics                                                                                   | Requirements |      |      | Method of Test, Ref to Cl |
|--------|---------------------------------------------------------------------------------------------------|--------------|------|------|---------------------------|
|        |                                                                                                   | FFP1         | FFP2 | FFP3 |                           |
| (1)    | (2)                                                                                               | (3)          | (4)  | (5)  | (6)                       |
| i)     | Sodium chloride test : Initial penetration of sodium chloride test aerosol @ 95 lpm, percent, Max | 20           | 6    | 3    | A-4.2 (see Note)          |
| ii)    | Paraffin oil test : Initial penetration of paraffin oil @ 95 lpm, percent, Max                    | —            | 2    | 1    | A-4.3 (see Note)          |

NOTE — A total of 12 filtering half masks shall be tested for each aerosol: 3 as received, 3 after temperature conditioning in accordance with A-1.2, 3 after the simulated wearing treatment in accordance with A-1.1 and 3 after the test for mechanical strength in accordance with A-1.3.

### 5.5 Compatibility with Skin

Materials that may come into contact with the wearer's skin shall not be known to have potential to cause irritation or any adverse effect to health.

### 5.6 Flammability

The material used shall not present a danger for the wearer and shall not be of highly flammable nature. When tested in accordance with A-5, the face piece shall not continue to burn after removal from the flame.

NOTE — It is not required that the filtering half mask still has to be usable after the test.

### 5.7 Carbon Dioxide Content of the Inhalation Air

When tested in accordance with A-6, the carbon dioxide content of the inhalation air (dead space) shall not exceed an average of 1.0 percent ( by volume).

### 5.8 Head Harness

5.8.1 The head harness shall be so designed that the filtering half mask can be donned and removed easily, when tested in accordance with A-2.

5.8.2 The head harness shall be adjustable or self-adjusting and shall hold the filtering half mask firmly and comfortably in position, and be capable of maintaining total inward leakage requirements for the device when tested in accordance with A-2 and A-3.

### 5.9 Field of Vision

The field of vision shall be acceptable when determined so during the practical performance tests in accordance with A-2. Comparative testing of the field of vision is carried out in accordance with the method described in A-8 of IS 14746.

### 5.10 Inhalation and Exhalation Valves

#### 5.10.1 Inhalation Valve(s)

A filtering half mask may have one or more inhalation valve(s).

#### 5.10.2 Exhalation Valve(s)

5.10.2.1 A filtering half mask may have one or more exhalation valve(s) and, if fitted, shall function correctly in all orientations, when tested according to A-3.

5.10.2.2 If an exhalation valve is provided, it shall be protected against dirt and mechanical damage, and shall be shrouded or shall include any other device that may be necessary for the filtering half mask to comply with A-3.

5.10.2.3 If an exhalation valve is provided, it shall continue to operate correctly after a continuous exhalation flow of 300 l/min over a period of 30 s. A total of 3 filtering half masks shall be tested; one as received and 2 temperature conditioned in accordance with A-1.2.

5.10.2.4 When the exhalation valve housing is attached to the face blank, it shall withstand axially a

tensile force of 10 N applied for 10 s, when tested in accordance with A-7.

### 5.11 Breathing Resistance

The breathing resistance of the valved and valveless filtering masks shall meet the requirements given in Table 2.

**Table 2 Breathing Resistance**

| Sl No. | Characteristics                                                                              | Requirements |            |            | Method of Test, Ref to Cl |
|--------|----------------------------------------------------------------------------------------------|--------------|------------|------------|---------------------------|
|        |                                                                                              | FFP1         | FFP2       | FFP3       |                           |
| (1)    | (2)                                                                                          | (3)          | (4)        | (5)        | (6)                       |
| 1      | Permitted resistance, mbar, <i>Max</i><br>– Inhalation @ 30 l/min<br>– Inhalation @ 95 l/min | 0.6<br>2.1   | 0.7<br>2.4 | 1.0<br>3.0 | A-8                       |
| 2      | Permitted resistance, mbar, <i>Max</i><br>– Exhalation @ 160 l/min                           | 3.0          | 3.0        | 3.0        | A-8                       |

1 mbar = 10<sup>5</sup> N/m<sup>2</sup> = 100 kPa; 1 bar = 10<sup>5</sup> Pa; 1 mbar = 100 kPa.

### 5.12 Clogging

**5.12.1** Two types of tests are available using coal dust for coal mining use and dolomite dust for other applications. Devices designed to be resistant to clogging (slow increase of breathing resistance when loaded with dust) shall be subjected to the treatment in accordance with A-9.

#### NOTES

1 Respirators shall be tested for clogging either with coal dust or dolomite depending on the end use.

2 For single use respirator, this test is optional.

### 5.12.2 Breathing Resistance After Clogging

#### 5.12.2.1 Valved filtering half masks

- The inhalation resistance shall not be greater than 4 mbar for FFP1 and 5 mbar for FFP2 when tested for breathing resistance at 95 l/min, in accordance with A-8 after the treatment.
- The exhalation resistance shall not be greater than 3 mbar at 160 l/min continuous flow.

#### 5.12.2.2 Valveless filtering half masks

The inhalation and exhalation resistances shall not be greater than 3 mbar for FFP1 and 4 mbar for FFP2, when tested for breathing resistance at 95 l/min continuous flow, in accordance with A-8.

### 5.12.3 Filter Penetration

All filtering half masks (valved and valveless) which meet the clogging requirement (*see* 5.12.1 and 5.12.2) shall also comply with the penetration requirements given in 5.4.2 after the treatment.

### 5.13 De-mountable Parts

All de-mountable parts (if fitted) shall be readily connected and secured, where possible by hand when tested according to A-11.

## 6 PACKING AND MARKING

### 6.1 Packing

The material shall be packed suitably as agreed to between the seller and the purchaser to prevent any damages during handling and transportation.

### 6.2 Marking

#### 6.2.1 Filtering Half Mask

Filtering half masks shall be clearly and durably marked with the following information:

- The name, trade-mark or other means of identification of the manufacturer;
- Class : FFP1, FFP2 or FFP3;
- The letter S (solid) or SL (solid and liquid) in accordance with filter penetration performance. These letters shall follow the class designation [*see* (b) above];
- If appropriate the letter D (dolomite) or C (coal) in accordance with clogging performance. These letters shall follow the class designation [*see* (b) above]; and
- Sub-assemblies and components with considerable bearing on safety shall be marked so that they can be identified.

NOTE — Colours do not represent a colour code, this means that colours are not related to protection provided.

**6.2.2** The package of the filtering half masks complying with this standard shall be clearly and durably marked with the following information:

- The name, trade-mark or other means of identification of the manufacturer;
- Class : FFP1, FFP2, FFP3;
- The year of manufacture plus the rated shelf life or the date of expiry of rated shelf life (where the reliable performance may be affected by ageing);
- The sentence 'see instructions for use'; and
- The package of filtering half masks which do not pass the paraffin oil test shall be clearly marked 'For use against solid aerosols only'.

NOTE — This may include water based aerosols (*see* 4.1).

#### 6.2.3 BIS Certification Marking

The filtering half masks may also be marked with the Standard Mark.

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**6.2.3.1** The use of the Standard Mark is governed by the provisions of the *Bureau of Indian Standards Act*, 1986 and the Rules and Regulations made thereunder. The details of conditions under which the licence for use of the Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

## 7 INSTRUCTIONS FOR USE

**7.1** Instructions for use shall accompany every filtering half mask.

**7.2** Instructions for use shall be in one or more languages acceptable to the country of application.

**7.3** The instructions for use of the equipment shall contain all information necessary for trained and qualified persons on the following:

- a) Application/limitation;
- b) The information only for single use, if applicable;
- c) State the suitability of the filters marked for solid aerosols only, against water based aerosols defined as: water based aerosols are those produced from solutions and/or suspensions of particulate materials in water such that the only workplace contaminant is attributed to this solid material;
- d) Checks prior to use;
- e) Donning, fitting;
- f) Use;
- g) Maintenance (preferably separately printed instructions); and
- h) Storage.

**7.4** The instructions shall be unambiguous if helpful, illustrations, part numbers, marking, etc, shall be added.

**7.5** Warning shall be given against problems likely to be encountered, for example, fit of filtering half mask (check prior to use); it is unlikely that the requirements for leakage will be achieved if facial, hair passes under the face seal; air quality (contaminants, oxygen deficiency); use of equipment in explosive atmosphere; and filtering half masks which do not pass the paraffin oil test shall be used against solid and water-based aerosols only.

**7.6** The instructions shall indicate that single-use filtering half masks shall be discarded after single-use.

## 8 SAMPLING AND CRITERIA FOR CONFORMITY

### 8.1 Lot

In a single consignment, all the half masks assembled

under uniform conditions of manufacture on the same day shall constitute a lot.

**8.1.1** Each lot shall be tested separately for ascertaining the conformity of the lot to the requirements of the specification. The number of half masks to be selected from the lot shall depend upon the size of the lot and shall be in accordance with Table 3.

**8.1.2** The masks shall be selected at random from the lot. For this purpose, reference may be made to IS 4905.

**Table 3 Number of Samples to be Tested from a Lot**  
(Clause 8.1.1)

| Sl No. | No. of Half Masks in the Lot | No. of Half Masks to be Selected in a Sample |
|--------|------------------------------|----------------------------------------------|
| (1)    | (2)                          | (3)                                          |
| i)     | Up to 50                     | 12                                           |
| ii)    | 51 to 150                    | 13                                           |
| iii)   | 151 and above                | 20                                           |

### 8.2 Number of Tests

**8.2.1** Each of the half masks selected from the lot according to col 2 of Table 3 shall be examined for visual inspection (5.13) and field of vision (5.9).

**8.2.1.1** The lot shall be considered to have satisfied the above requirements if none of the masks in the sample fails. Otherwise, the lot shall be rejected.

**8.2.2** From the samples having been found satisfactory as per 8.2.1, required number of samples according to col 3 of Table 3 shall be further tested for the requirements of carbon dioxide content (5.7), breathing resistance (5.11), practical performance test (5.3), and cleaning and disinfection (5.2) in this sequence.

**8.2.2.1** The lot shall be considered to have satisfied the above requirements if there is no failure in the sample. Otherwise, the lot shall be rejected.

**8.2.3** Required number of masks, out of the sample already been found satisfactory as per 8.2.2 shall be drawn and made into two parts.

**8.2.3.1** The samples under one part shall be conditioned as per A-1 and shall be tested for leakage of half masks (5.4), flammability (5.6), and inhalation and exhalation valve (5.10), if fitted, and clogging.

**8.2.3.2** The samples under other part shall be tested for the requirements of leakage (5.4), flammability (5.6), head harness tests (5.8), exhalation valves (5.10.2), performance tests (5.3) and exhalation valves housing (5.10.2.4).

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**8.2.3.3** The lot shall be considered as conforming to the requirements of this specification if all the half masks pass the requirements specified in **8.2.3.1**

and **8.2.3.2**. Otherwise the lot shall be rejected.

**8.3** A summary of tests applicable for half masks is given in Table 4.

**Table 4 Summary of Tests**

| Sl No.<br>(1) | No. of Samples<br>(2) | Test Criteria<br>(3)                                                           | Conditioning (Yes/No)<br>(4)                 | Clause<br>(5)                      |
|---------------|-----------------------|--------------------------------------------------------------------------------|----------------------------------------------|------------------------------------|
| i)            | All                   | Visual inspection                                                              | A.R.                                         | A-11                               |
| ii)           | 5                     | Cleaning and disinfection                                                      | A.R.                                         | As per manufacturer's instructions |
| iii)          | 4                     | Flammability                                                                   | A.R. (2)<br>T.C. (2)                         |                                    |
| iv)           | 3                     | Carbon dioxide content                                                         | A.R.                                         | A-6                                |
| v)            | 3                     | Exhalation valve pull                                                          | A.R. (1)<br>S.W. (1)<br>T.C. (1)             | A-7                                |
| vi)           | 3                     | Exhalation valve flow                                                          | A.R. (1)<br>T.C. (2)                         | —                                  |
| vii)          | 9                     | Breathing resistance                                                           | A.R. (3)<br>S.W. (3)<br>T.C. (3)             | A-8                                |
| viii)         | 12 (for each aerosol) | Particle filter penetration                                                    | A.R. (3)<br>S.W. (3)<br>T.C. (3)<br>M.S. (3) | A-4                                |
| ix)           | 10                    | Total inward leakage                                                           | A.R. (5)<br>T.C. (5)                         | A-3                                |
| x)            | 6                     | Clogging test (Optional for FFP1 + FFP2 single use devices only) <sup>1)</sup> | A.R. (2)<br>T.C. (2)<br>M.S. (2)             | A-9                                |
| xi)           | 2                     | Practical performance                                                          | A.R.                                         | A-2                                |

A.R. As received  
M.S. Mechanical strength  
S.W. Simulated wearing treatment  
T.C. Temperature conditioned

<sup>1)</sup> Two tests available:  
a) Coal dust: for mining applications  
b) Dolomite dust: for other applications

## ANNEX A

(Clauses 5.1.1, 5.1.2, 5.3, 5.4.1.1, 5.4.2, 5.6, 5.7, 5.8.1, 5.8.2, 5.9, 5.10.2.1, 5.10.2.2, 5.10.2.3, 5.10.2.4, 5.11, 5.12.1, 5.12.2.1, 5.12.2.2, 5.12.3 and 5.13)

### METHODS OF TESTS FOR FILTERING HALF MASKS

#### A-1 CONDITIONING

##### A-1.1 Simulated Wearing Treatment

Conditioning by simulated wearing treatment shall be carried out by the following process.

A breathing machine is adjusted to 25 cycles/min and 2l/stroke. The filtering half mask is mounted on a Sheffield dummy head. For testing, a saturator is incorporated in the exhalation line between the

breathing machine and the dummy head, the saturator being set in a temperature in excess of 37°C to allow for cooling of the air before it reaches the mouth of the dummy head. The air shall be saturated at 37 ± 2°C at the mouth of the dummy head. In order to prevent excess water spilling out of the dummy's mouth and contaminating the filtering half mask, the head shall be inclined so that the water runs away from the mouth and is collected in a trap.

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The breathing machine is brought into operation, the saturator switched on and the apparatus allowed to stabilize. The filtering half mask under the test shall then be mounted on the dummy head. During the test time at approximately 20 min intervals, the filtering half mask shall be completely removed from the dummy head and refitted such that during the test period, it is fitted ten times to the dummy head.

### A-1.2 Temperature Conditioning

The filtering half masks shall be exposed to the following thermal cycle:

- for 24 h to a dry atmosphere of  $70 \pm 3^\circ\text{C}$ ,
- for 24 h to a temperature of  $-30 \pm 3^\circ\text{C}$ , and

allowed to return to room temperature for at least 4 h between exposures and prior to subsequent testing.

### A-1.3 Mechanical Strength

Before testing for breathing resistance, filtration efficiency and clogging, the filter shall be subjected to a test in accordance with A-10 simulating rough usage of filter. After this treatment, the filters shall show no mechanical defects and shall meet the requirements for breathing resistance, filtration efficiency and clogging.

## A-2 PRACTICAL PERFORMANCE TEST

A total of 2 filtering half masks shall be tested, both as received. All tests shall be carried out by two test subjects at ambient temperature, and the test temperature and humidity shall be recorded. Prior to the test, there shall be an examination as per A-11 to assure that the filtering half mask is in good working condition and that it can be used without hazard. For the test, persons shall be selected who are familiar with using such or similar equipment. During the tests, the filtering half mask shall be subjectively assessed by the wearer and after the test, comments on the following shall be recorded:

- Head harness comfort,
- Security of fastenings,
- Field of vision, and
- Any other comments reported by the wearer on request.

### A-2.1 Walking Test

The subjects wearing normal working clothes and wearing the filtering half mask shall walk at a regular rate of 6 km/h on a level course. The test shall be continuous without removal of the filtering half mask for a period of 10 min.

### A-2.2 Work Simulation Test

The filtering half mask shall be tested under conditions

which can be expected during normal use. During this test, the following activities shall be carried out in simulation of the practical use of the filtering half mask. The test shall be completed within a total working time of 20 min.

The sequence of activities is at the discretion of the test station. The individual activities shall be arranged so that sufficient time is left for the comments prescribed.

- Walking on the level with headroom of  $1.3 \pm 0.2$  m for 5 min.
- Crawling on the level with head room of  $0.7 \pm 0.05$  m for 5 min.
- Filling a small basket (*see* Fig. 1, approximate volume = 8 l) with rubber chippings or other suitable material from a hopper which stands 1.5 m high and has an opening at the bottom to allow the contents to be shovelled out and a further opening at the top where the basket full of rubber chippings shall be returned.

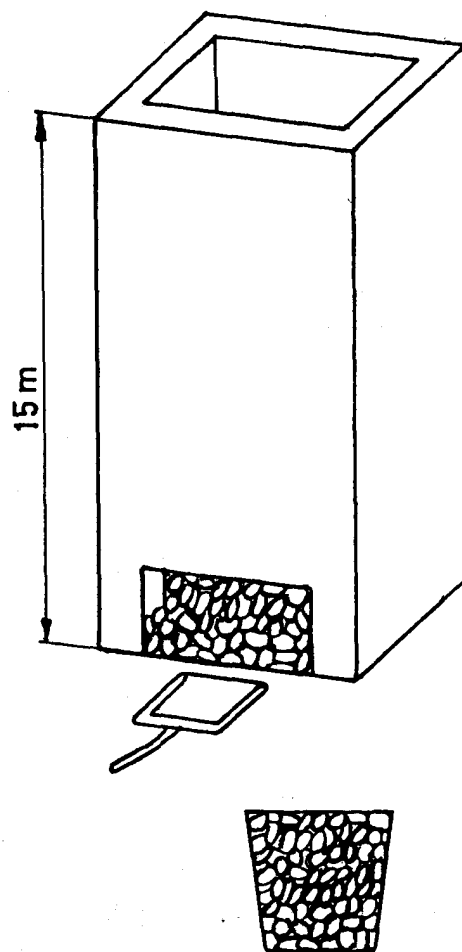


FIG. 1 BASKET AND HOPPER, RUBBER CHIPPINGS



The subject shall stoop or kneel as he wishes and fill the basket with rubber chippings. He shall then lift the basket and empty the contents back into the hopper. This shall be done 20 times in 10 min.

### A-3 LEAKAGE

#### A-3.1 General Test Procedure

##### A-3.1.1 Total Inward Leakage

A total of 10 test specimens shall be tested; 5 as received and 5 after temperature conditioning in accordance with A-1.2. The inward leakage shall be tested using a solid aerosol. The reference method is the sodium chloride aerosol test. Prior to the test, there shall be an examination as per A-11 to ensure that the filtering half mask is in good working condition and that it can be used without hazard.

For the test, persons shall be selected who are familiar with using such or similar equipment. A panel of ten clean-shaven persons (without beards or sideburns) shall be selected covering the spectrum of facial characteristics of typical users (excluding significant abnormalities). It is to be expected that exceptionally some persons cannot be satisfactorily fitted with a filtering half mask. Such exceptional subjects shall not be used for testing filtering half masks.

In the test report, the faces of the ten test subjects shall be described (for information only) by the four facial dimensions (in mm) illustrated in Fig. 2.

##### A-3.1.2 Apparatus

###### A-3.1.2.1 Treadmill

A level treadmill is required which is capable of working at 6 km/h.

##### A-3.1.3 Test Atmosphere

The test atmosphere shall preferably enter the top of

the enclosure through a flow distributor and be directed downwards over the head of the test subject at a minimum flow rate of 0.12 m/s. The concentration of the test agent inside the effective working volume shall be checked to be homogeneous. The flow rate should be measured close to the subject's head.

##### A-3.1.4 Test Procedure

The test subjects shall be asked to read to manufacturer's fitting instructions and if more than one size of filtering half mask is manufactured, the test subject shall be asked to select the size deemed by him to be the most appropriate. If necessary, the subjects shall be shown by the test supervisor how to fit the filtering half mask correctly in accordance with the fitting instructions.

The test subjects shall be informed that if they wish to adjust the filtering half mask during the test they may do so. However, if this is done, the relevant section of the test shall be repeated having allowed the system to re-settle.

The test subjects shall have no indication of the results as the test proceeds. After fitting the filtering half mask each test subject shall be asked 'Does the mask fit'? If the answer is 'Yes', continue the test. If the answer is 'No', take the test subject off the panel, report the fact and replace with another test subject. The test sequence shall be as follows:

- Ensure the test atmosphere is OFF.
- Place the test subject in the enclosure. Connect up the face piece sampling probe. Have the test subject walk at 6 km/h for 2 min. Measure the test agent concentration inside the filtering half mask to establish the background level.
- Obtain a stable reading.
- Turn the test atmosphere ON.

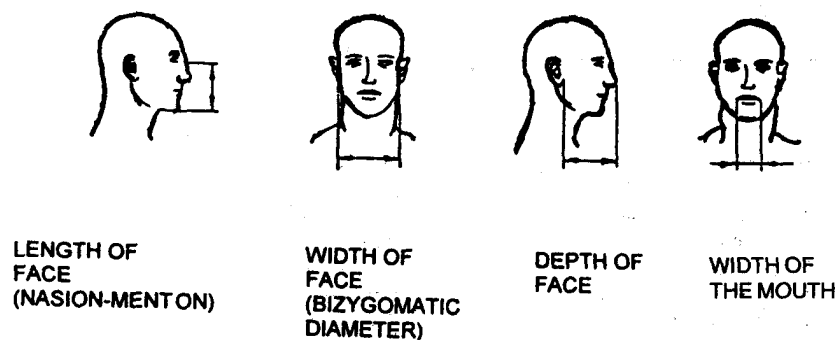


FIG. 2 FACIAL DIMENSIONS

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- e) The subject shall continue to walk for a further 2 min or until the test atmosphere has stabilized.
- f) Whilst still walking the subject shall perform following exercises:
  - 1) Walking for 2 min without head movement or talking.
  - 2) Turning head from side to side (approximately 15 times), as if inspecting the walls of a tunnel for 2 min.
  - 3) Moving the head up and down (approximately 15 times), as if inspecting the roof and floor for 2 min.
  - 4) Reciting the alphabet or an agreed text out loud as if communicating with a colleague for 2 min.
  - 5) Walking for 2 min without head movement or talking.
- g) Turn off the test atmosphere and when the test agent has cleared from the enclosure, remove the subject. After each test the filtering half mask shall be replaced by a new sample.

### 3.1.5 Results

Record the following as test results:

- a) Enclosure concentration.
- b) The leakage over each exercise period.

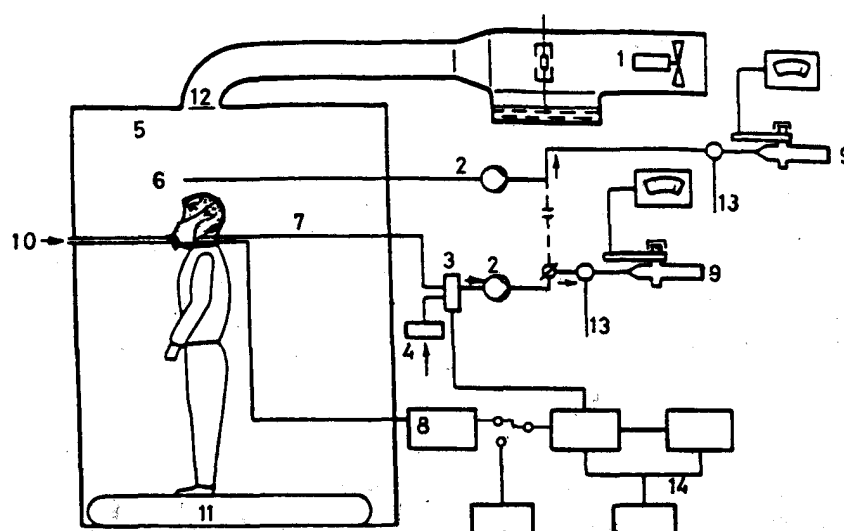
### A-3.2 Sodium Chloride (NaCl) — Method

#### A-3.2.1 Principle

The subject wearing the filtering half mask under test walks on a treadmill over which is an enclosure. Through this enclosure flows a constant concentration of NaCl aerosol. The air inside the filtering half mask is sampled and analyzed during the inhalation phase of the respiratory cycle to determine the NaCl content. The sample is extracted by punching a hole in the filtering half mask and inserting a probe through which the sample is drawn. The pressure variation inside the filtering half mask is used to actuate a change-over valve so that inhaled air only is sampled. A second probe is inserted for this purpose.

#### A-3.2.2 Apparatus

##### A-3.2.2.1 Test equipment (see Fig. 3)



- |   |                   |    |                           |
|---|-------------------|----|---------------------------|
| 1 | Atomizer          | 8  | Manometer                 |
| 2 | Pump              | 9  | Photometer                |
| 3 | Change-over valve | 10 | Filtering half mask       |
| 4 | Filter            | 11 | Treadmill                 |
| 5 | Enclosure         | 12 | Ducting and baffle        |
| 6 | Enclosure sample  | 13 | Additional air            |
| 7 | Mask sample       | 14 | Pulsed sampling interface |

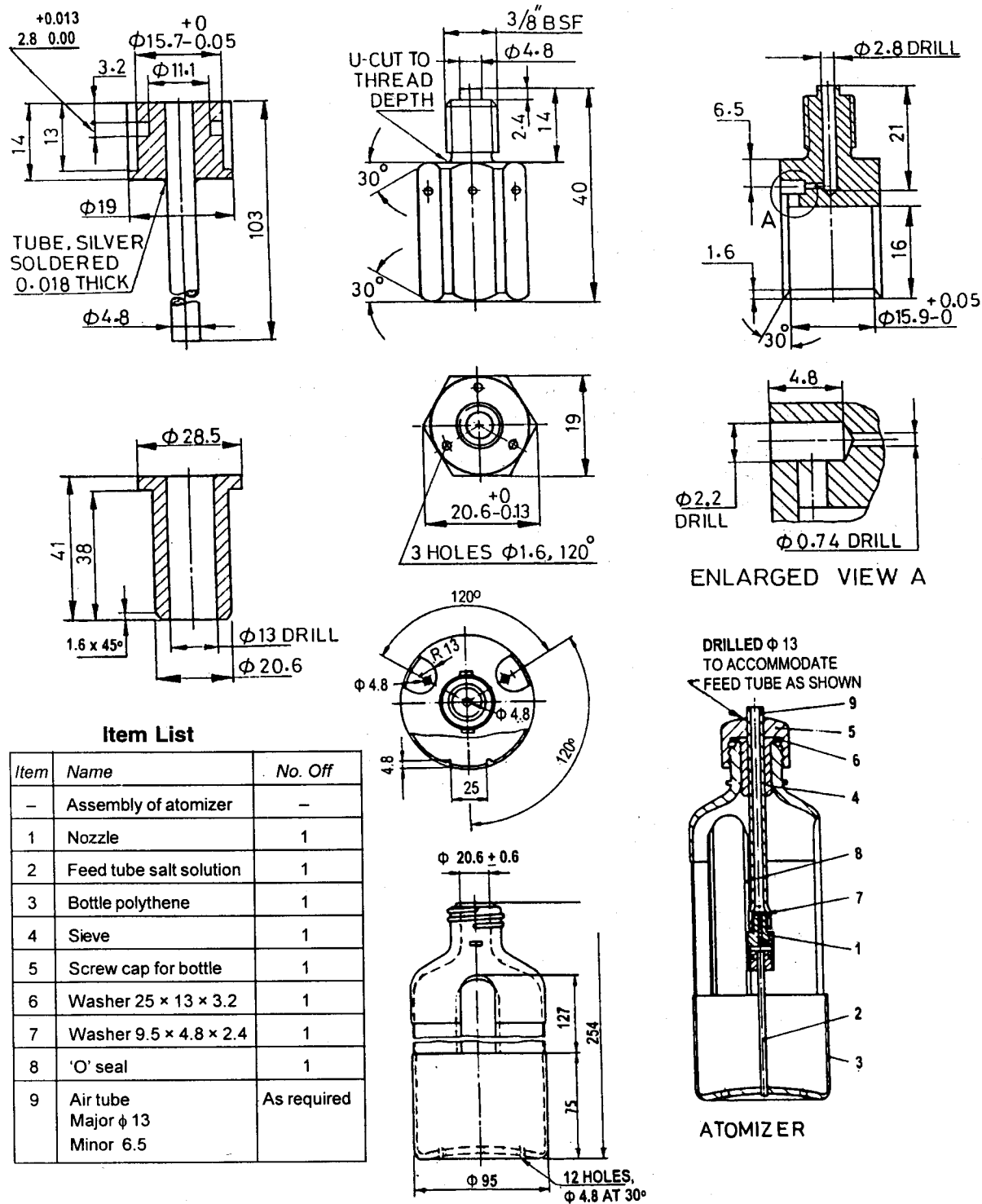
FIG. 3 TYPICAL APPARATUS USED IN THE DETERMINATION OF INWARD LEAKAGE USING SODIUM CHLORIDE

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### A-3.2.2.2 Aerosol generator

The NaCl aerosol shall be generated from a 2 percent solution of reagent grade NaCl in distilled water. An atomizer equivalent to the type described should be used (see Fig. 4). This requires an air flow rate of

100 l/min at a pressure of 7 bar. The atomizer and its housing shall be fitted into a duct through which a constant flow of air is maintained. It may be necessary to heat or dehumidify the air in order to obtain complete drying of the aerosol particles.



| Item List |                                          |             |
|-----------|------------------------------------------|-------------|
| Item      | Name                                     | No. Off     |
| -         | Assembly of atomizer                     | -           |
| 1         | Nozzle                                   | 1           |
| 2         | Feed tube salt solution                  | 1           |
| 3         | Bottle polythene                         | 1           |
| 4         | Sieve                                    | 1           |
| 5         | Screw cap for bottle                     | 1           |
| 6         | Washer 25 x 13 x 3.2                     | 1           |
| 7         | Washer 9.5 x 4.8 x 2.4                   | 1           |
| 8         | 'O' seal                                 | 1           |
| 9         | Air tube<br>Major $\phi 13$<br>Minor 6.5 | As required |

NOTE — All burns and sharp edges to be removed. MAT. Stainless steel.

All dimensions in millimetres.

FIG. 4 ASSEMBLY OF ATOMIZER

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### A-3.2.2.3 Test agent

The mean NaCl concentration within the enclosure shall be  $8 \pm 4 \text{ mg/m}^3$  and the variation throughout the effective working volume shall be not more than 10 percent. The particle size distribution shall be 0.02 to  $2 \mu\text{m}$  equivalent aerodynamic diameter with a mass median diameter of  $0.6 \mu\text{m}$ .

### A-3.2.2.4 Flame photometer

A flame photometer shall be used to measure the concentration of NaCl inside the filtering half mask. Essential performance characteristics for a suitable instrument are as follows:

- It should be a flame photometer specifically designed for the direct analysis of NaCl aerosol.
- It should be capable of measuring concentrations of NaCl aerosol between  $15 \text{ mg/m}^3$  and  $5 \text{ mg/m}^3$ .
- The total aerosol sample required by the photometer should not be greater than 15 l/min.
- The response time of the photometer, excluding the sampling system, should not be greater than 500 milliseconds.
- It is necessary to reduce the response to other elements, particularly carbon, the concentration of which will vary during the breathing cycle. This will be achieved by ensuring that the band pass width of the interference filter is not greater than 3 nm and that all necessary side band filters are included.

### A-3.2.2.5 Sample selector

A system is required which will switch the sample to the photometer only during the inhalation phase of the respiratory cycle. During the exhalation phase, clean air shall be fed to the photometer. The essential elements of such a system are as follows:

- An electrically operated valve with a response time of the order of 100 ms. The valve should have the minimum possible dead space compatible with straight-through, unrestricted flow when open.
- A pressure sensor which is capable of detecting a minimum pressure change of approximately 0.05 mbar and which can be connected to a probe inserted in the cavity of the filtering half mask. The sensor shall have an adjustable threshold and be capable of differential signalling when the threshold is crossed in either direction. The sensor shall work reliably when subjected to the accelerations produced by the head movements of the subject.

- An interfacing system to actuate the valve in response to a signal from the pressure sensor.
- A timing device to record the proportion of the total respiratory cycle during which sampling took place.

### A-3.2.2.6 Sampling probe

The probe shall be fitted securely in an air-tight manner to the filtering half mask as near as possible to the centre line of the filtering half mask. A multiple hole sampling probe is strongly recommended. Measures shall be taken to prevent the influence of condensation in the sampling probe on the measurement (by supplying dry air). Figure 5 shows a design that has been found suitable. The probe is adjusted so that it just touches the wearer's lips.

### A-3.2.2.7 Sample pump

If no pump is incorporated into the photometer, an adjustable flow pump is used to withdraw an air sample from the filtering half mask under test. This pump is so adjusted as to withdraw a constant flow of 1 l/min from the sample probe. Dependent on the type of photometer, it may be necessary to dilute the sample with clean air.

### A-3.2.2.8 Sampling of enclosure concentration

The aerosol enclosure concentration is monitored during the tests using a separate sampling system to avoid contamination of the filtering half mask sampling lines. It is preferable to use a separate flame photometer for this purpose.

If a second photometer is not available, sampling of the enclosure concentration using a separate sampling system and the same photometer may be made. However, time will then be required to allow the photometer to return to a clean background.

### A-3.2.2.9 Pressure detection probe

A second probe is fitted near to the sample probe and is connected to the pressure sensor.

### A-3.2.3 Expression of Results

The leakage  $P$  shall be calculated from the measurements made over the last 100 s of each of the exercise periods to avoid carry over of results from one exercise to another.

$$P = \frac{C_2 X}{C_1} \times \frac{t_{\text{IN}} + t_{\text{EX}}}{t_{\text{IN}}} \times 100$$

where

- $C_1$  = Challenge concentration,
- $C_2$  = Measured mean concentration,
- $t_{\text{IN}}$  = Total duration of inhalation,
- $t_{\text{EX}}$  = total duration of exhalation, and
- $P$  = leakage in percent.

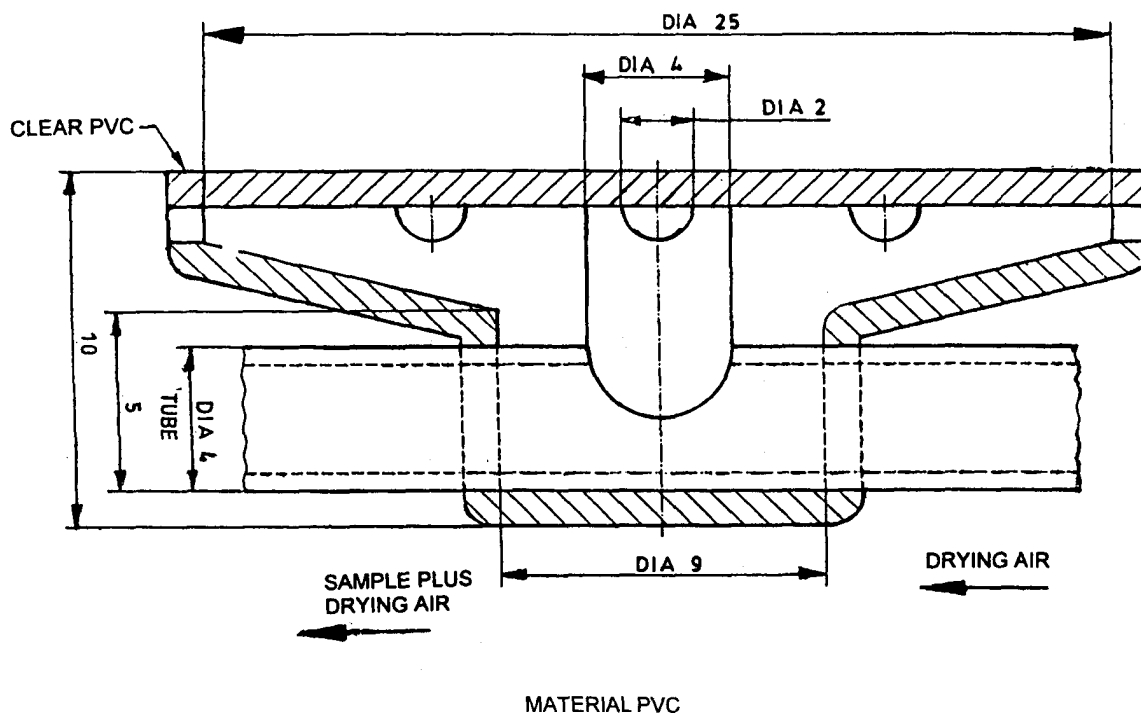


FIG. 5 SAMPLE PROBE

Measurement of  $C_2$  is preferably made using an integrating recorder.

#### A-4 FILTRATION EFFICIENCY

A-4.1 The filters shall be tested before and after the following temperature treatment. The filters shall be exposed for:

- 24 h to dry atmosphere of  $70 \pm 3^\circ\text{C}$ , and
- 24 h to temperature of  $-30 \pm 3^\circ\text{C}$ .

The methods used for testing filters against solid and liquid aerosols are:

- sodium chloride test according to A-4.2, and
- paraffin oil test according to A-4.3.

Sodium chloride test according to A-4.2 is the only method used for testing filters against solid aerosols.

Each test shall be made with 3 specimens.

##### A-4.1 Sodium Chloride Test

An aerosol of sodium chloride particles is generated by atomizing an aqueous solution of the salt and evaporating the water. The concentration of this aerosol is measured before and after the filter under the test by means of flame photometry. Accurate determinations are possible in the range  $< 0.000\ 01$  to 100 percent filter penetration.

##### A-4.2.1 Apparatus

The apparatus is shown in Fig. 6. The aerosol is generated using a Collison atomizer filled with 1 percent solution of sodium chloride. The atomizer, which is shown in Fig. 7, consists of a glass reservoir into which is sealed an atomizer head having three spray nozzles. Air is supplied to the atomizer at a pressure of 3.45 bar and the resulting liquid spray impinges on a baffle which removes the large particles. The particles which do not impact are removed in the air flow and, on mixing with dry air, the water evaporates leaving a dry sodium chloride aerosol.

The aerosol produced by this method is polydisperse with a mass median particle diameter of approximately  $0.6\ \mu\text{m}$ . The particle size distribution is given in Fig. 8. It has been found that the aerosol remains constant within acceptable limits with respect to particle size and concentration provided that the supply pressure is in the range of 3.31-3.59 bar and the flow rate of air to the three nozzles is 12.5-13.0 l/min. The output is mixed with 82 l/min of dry air giving a total flow of 95 l/min.

The salt solution in the atomizer is consumed at a rate of approximately 15 ml/h. This loss is due in part to the atomization of the solution and in part to evaporation of water from the reservoir. The volume of the reservoir is such that the change in concentration and loss in volume of the solution during an 8-h period

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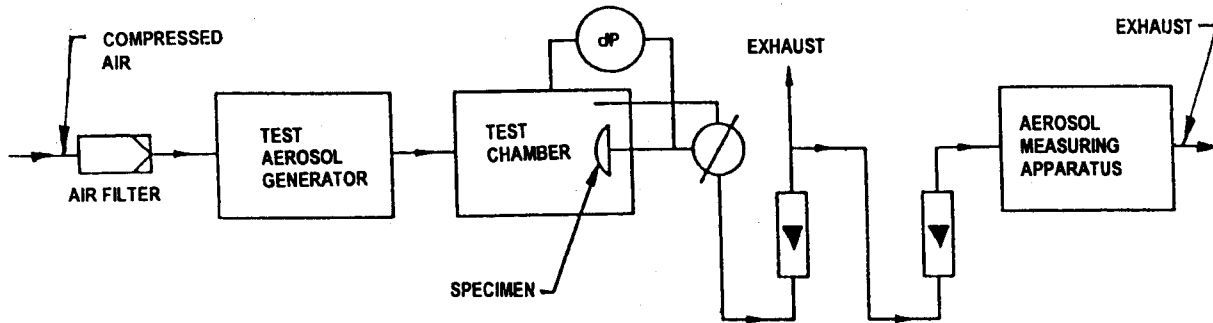
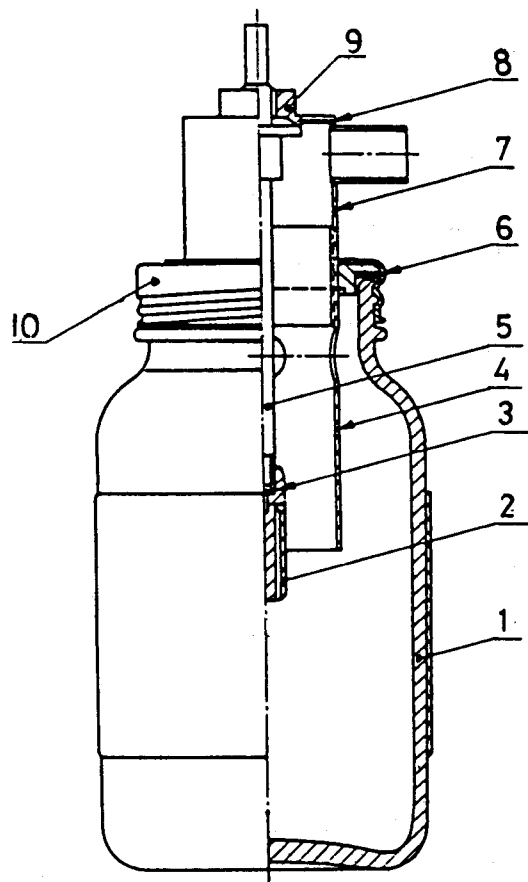


FIG. 6 APPARATUS FOR SODIUM CHLORIDE TEST



- 1 Glass screw top jar
- 2 Nozzle
- 3 Fibre washer  
4.5 mm external diameter  
2.0 mm internal diameter  
0.8 mm thickness
- 4 Sleeve
- 5 Stem
- 6 Rubber gasket
- 7 Head
- 8 Rubber gasket  
25.0 mm external diameter  
10.0 mm internal diameter  
1.5 mm thickness  
Seals to be airtight
- 9 Nut
- 10 Screw cap

FIG. 7 ATOMIZER

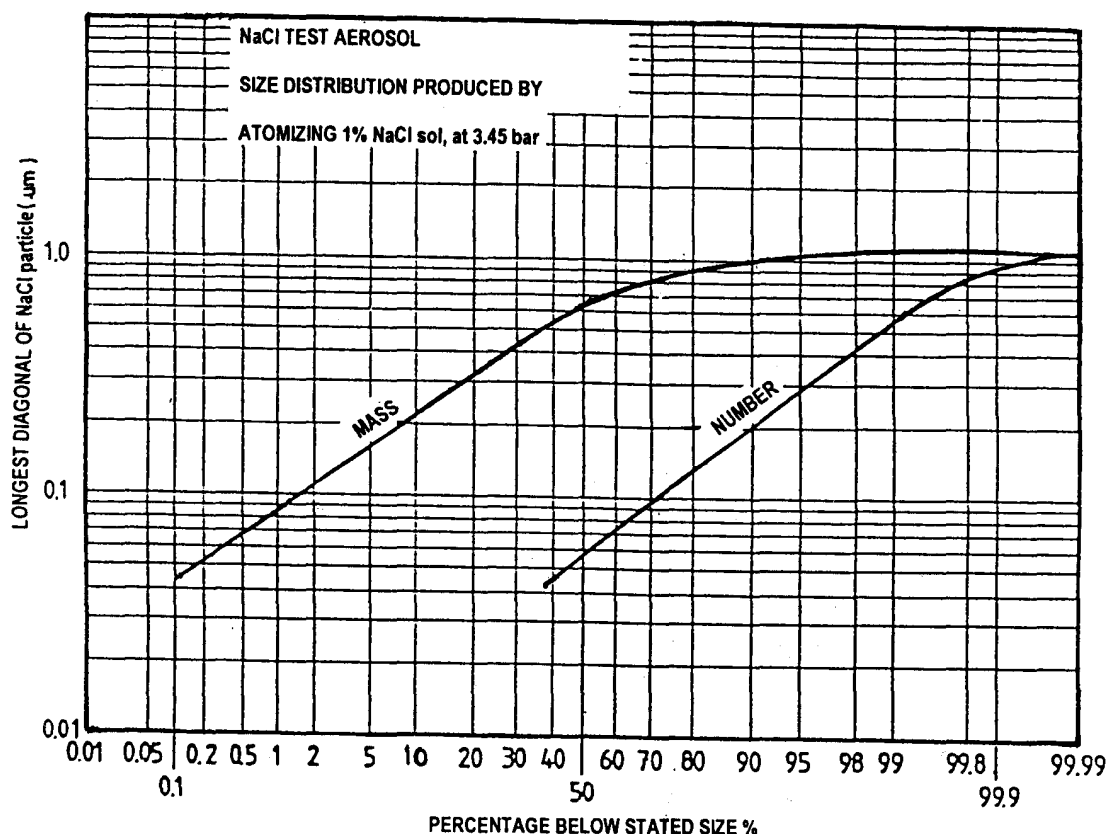


FIG. 8 PARTICLE SIZE DISTRIBUTION

will not cause an appreciable change in the characteristics of the test aerosol.

The sodium chloride aerosol is analyzed before and after the filter under test by flame photometry. The photometer used for this analysis can be any suitable instrument having the required sensitivity, however, a photometer specially designed to meet these requirements is available. The instrument is hydrogen flame photometer. The hydrogen burner is housed in a vertical flame tube which opens at its lower end into the sample tube through which the aerosol to be analyzed flows. The flow of aerosol to the flame is controlled by convection and is held constant with a bleed valve.

A small quantity of filtered air is fed continuously into the sample tube downstream of the inlet to the flame tube. The function of this supply is to prevent room air, which may contain considerable quantities of sodium salts, from reaching the burner when there is no flow through the sample tube.

The hydrogen burner, which gives a flame symmetrical about the vertical axis, is surrounded by a heat-proof glass tube. This tube has to be optically homogeneous to minimize the effect on the light transmitted by the flame.

Sodium chloride particles in air passing through the flame tube are vapourized giving the characteristic sodium emission at 589 nm. The intensity of this emission is proportional to the concentration of sodium in the air flow.

The intensity of the light emitted by the flame is measured using a photomultiplier tube. To separate the sodium emission from background light of other wavelengths, a narrow band interference filter with appropriate sideband filters is used. This filter should preferably have a half-peak band width of not more than 5 nm.

As the photomultiplier output is only proportional to the incident light over a relatively small range, high light intensities are attenuated by neutral density filters. These filters are accurately calibrated in conjunction with the interference filter in use and so the actual light intensity can be calculated from the output of the photomultiplier. The signal from the photomultiplier is amplified and displayed on a meter or chart recorder.

Calibration of the flame photometer will depend on the detailed design of the instrument and the manufacturer's instructions should be followed if

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reliable results are to be obtained. In general, however, the methods which may be used are multiple dilution of the aerosol dilution of the atomizer solution or a combination of both. If aerosol or solutions dilution is used alone, the lower calibration limit is approximately two orders of magnitude higher than the ultimate sensitivity of the instrument.

Where a photomultiplier with attenuating filters is used for detection this is unimportant as the photomultiplier measures a constant range of light levels over the entire range for the instrument and the values of the attenuating filters are known and invariable. Hence, the calibration curve is linear at low concentrations and can safely be extrapolated to the lower values. The upper limit of linearity of the calibration curve is approximately  $0.12 \text{ mg/m}^3$  due to re-absorption of light within the flame. Non-linear calibration is possible above this point up to approximately  $15 \text{ mg/m}^3$ . Where other detectors are used, this may not be the case and a combination technique would be required to reach the ultimate sensitivity.

### A-4.2.2 Test Conditions

Particle size distribution of the test aerosol (see Fig. 8).

|                                     |                                  |
|-------------------------------------|----------------------------------|
| Flow rate of test aerosol           | : 95 l/min                       |
| Aerosol concentration               | : $8 \pm 4 \text{ mg/m}^3$       |
| Air pressure to atomizer            | : $3.45 \pm 0.14 \text{ bar}$    |
| Flow rate to atomizer               | : $12.75 \pm 0.25 \text{ l/min}$ |
| Flow rate of diluting air           | : 82 l/min                       |
| Flow rate of hydrogen to photometer | : 450-500 ml/min                 |
| Wavelength of sodium emission       | : 589 nm                         |
| Air temperature                     | : ambient                        |
| Relative humidity                   | : < 60 percent                   |

### A-4.2.3 Test Procedure

The test aerosol is fed into the test chamber, where the filter under test is fixed. A flow of 95 l/min is blown through the filter and the aerosol is fed into the test chamber, where the filter under test is fixed. A flow of 95 l/min is blown through the filter and the aerosol concentration is measured immediately before and after the filter by the photometer. The initial penetration shall be measured at  $3 \pm 0.5 \text{ min}$ .

### A-4.2.4 Calculation of the Penetration

$$P = \frac{C_2}{C_1} \times 100 \text{ percent}$$

where

$P$  = Penetration

$C_1$  = Sodium chloride concentration before the filter, and

$C_2$  = Sodium chloride concentration after the filter.

### A-4.3 Paraffin Oil Test

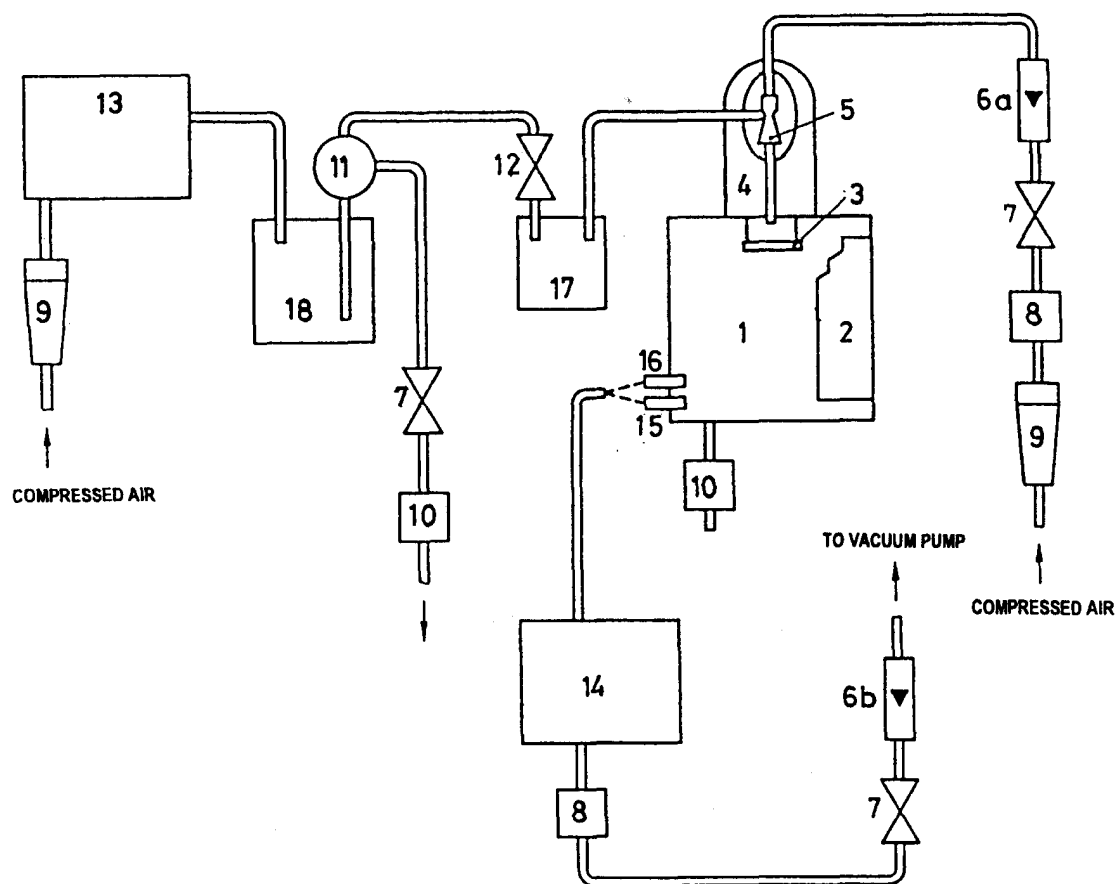
An aerosol of paraffin oil droplets is generated by atomizing heated paraffin oil. The concentration of this aerosol is measured before and after the filter under test by means of an aerosol photometer. Accurate determination are possible in the range  $< 0.0003$  to 100 percent filter penetration.

#### A-4.3.1 Apparatus

The apparatus is shown in Fig. 9. The aerosol is generated using an atomizer (Fig. 10 and 11). The atomization vessel (6) is filled with paraffin oil (paraffinum perliquidum CP 27 DAB 7), so that the oil level is between the minimum/maximum-marks (10). The atomization vessel is heated by means of an electric heating device (8), so that the temperature of the oil is kept at  $100^\circ\text{C}$  by means of a thermostat (9). The temperature is measured by the thermometer (11). Filtered compressed air at 4 bar (3, 4) is pre-heated (8) and blown through the multiple nozzle unit (12 and Fig. 11). Large droplets in the generated oil mist are separated in the control nozzle (13) and in the spiral tube (15). In the mixing vessel (5), the oil droplets and oil vapour are diluted with 50 l/min filtered air, measured by the flowmeter (2). Since the diluting air is at room temperature, the oil vapour condenses in the mixing vessel. The generated aerosol is the test aerosol, which is reduced to the test concentration of  $(20 \pm 5) \text{ mg/m}^3$  by wasting an appropriate fraction of the oil mist (see Fig. 9, item 18 in connection with items 11, 7, 10, 12 and 17) and by further dilution with filtered air at a flow rate of 83 l/min in the blowers actuated by air power (type Friedrichs-Antilinger, see Fig. 9, item 5 and Fig. 12). The test aerosol produced by this method is polydisperse. The particle size distribution is a logarithmic normal distribution with the median Stokes diameter of  $0.4 \mu\text{m}$  (for the number distribution) and the logarithmic standard deviation 0.26 (see Fig. 13).

The test aerosol is fed into the test chamber [Fig. 9, (1)], where the filter under test is to be fixed (15). The excess of the aerosol is filtered by a high efficiency filter with a low flow resistance (10). A flow rate of 95 l/min is drawn through the filter under test. The test concentration is measured before and after the filter under test by means of an integrating light scattering photometer. The principle of the aerosol photometer is shown in Fig. 14. The instrument is a  $45^\circ$  scattering photometer. The light source is directed to the measuring cell and to the photomultiplier. The direct beam to the multiplier is interrupted by a chopper, so that the scattered light from the particles is always corrected for the source intensity variations. The





- |                                                                                                                 |                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Test chamber: Rigid transparent material, diameter 500 mm, height 500 mm, covered with plywood on both sides. | 10 High efficiency filter with low resistance.                                                                                                                                                                                                                                                                       |
| 2 Tightly fitting door of the chamber.                                                                          | 11 Tee for taking out the amount of oil mist necessary for testing.                                                                                                                                                                                                                                                  |
| 3 Dish for collecting oil running down the walls of the tube.                                                   | 12 Needle valve controlling the oil mist concentration in the chamber.                                                                                                                                                                                                                                               |
| 4 Cover for the blowers actuated by air power.                                                                  | 13 Oil mist generator.                                                                                                                                                                                                                                                                                               |
| 5 Blowers actuated by air power for taking the concentrated oil mist into the chamber.                          | 14 Aerosol photometer.                                                                                                                                                                                                                                                                                               |
| 6 Flowmeters, range 800–8 000 l/h                                                                               | 15 Connecting pipe to test object.                                                                                                                                                                                                                                                                                   |
| a) for measuring the propellant air for the blowers (5 000 l/h), and                                            | 16 Probe for measuring the oil mist concentration in the chamber. The aerosol photometer is connected with 15 or 16 as required by means of a short tube, the connecting pipe not employed is to be closed tightly. The tubes for the oil mist are textile reinforced plastic tubes with an inner diameter of 19 mm. |
| b) for measuring the test flow rate (95 l/min).                                                                 | 17 Woulfe's bottle.                                                                                                                                                                                                                                                                                                  |
| 7 Valves controlling the flow rate.                                                                             | 18 Buffer volume of 5 l.                                                                                                                                                                                                                                                                                             |
| 8 High efficiency filters.                                                                                      |                                                                                                                                                                                                                                                                                                                      |
| 9 Reducing valves, range 1-5 bar at a pre-pressure of 6-10 bar.                                                 |                                                                                                                                                                                                                                                                                                                      |

FIG. 9 APPARATUS FOR PARAFFIN OIL TEST

reference beam is attenuated by means of neutral density filters and of a neutral density wedge automatically to the intensity of the scattered light beam.

The intensity of the scattered light, which is a measure for the aerosol concentration, is displayed.

#### A-4.3.2 Test Conditions

- Particle size distribution of the test aerosol (see Fig. 13).
- Flow rate through the filter under test : 95 l/min

- Aerosol concentration :  $20 \pm 5 \text{ mg/m}^3$
- Air temperature : ambient
- Air pressure to atomizer :  $4 \pm 0.15 \text{ bar}$
- Flow rate to atomizer :  $13.5 \pm 0.5 \text{ l/min}$
- Mixing air flow rate in aerosol generator : 50 l/min
- Flow rate of diluting air : 83 l/min
- Temperature of the oil : between 100 and 110 °C

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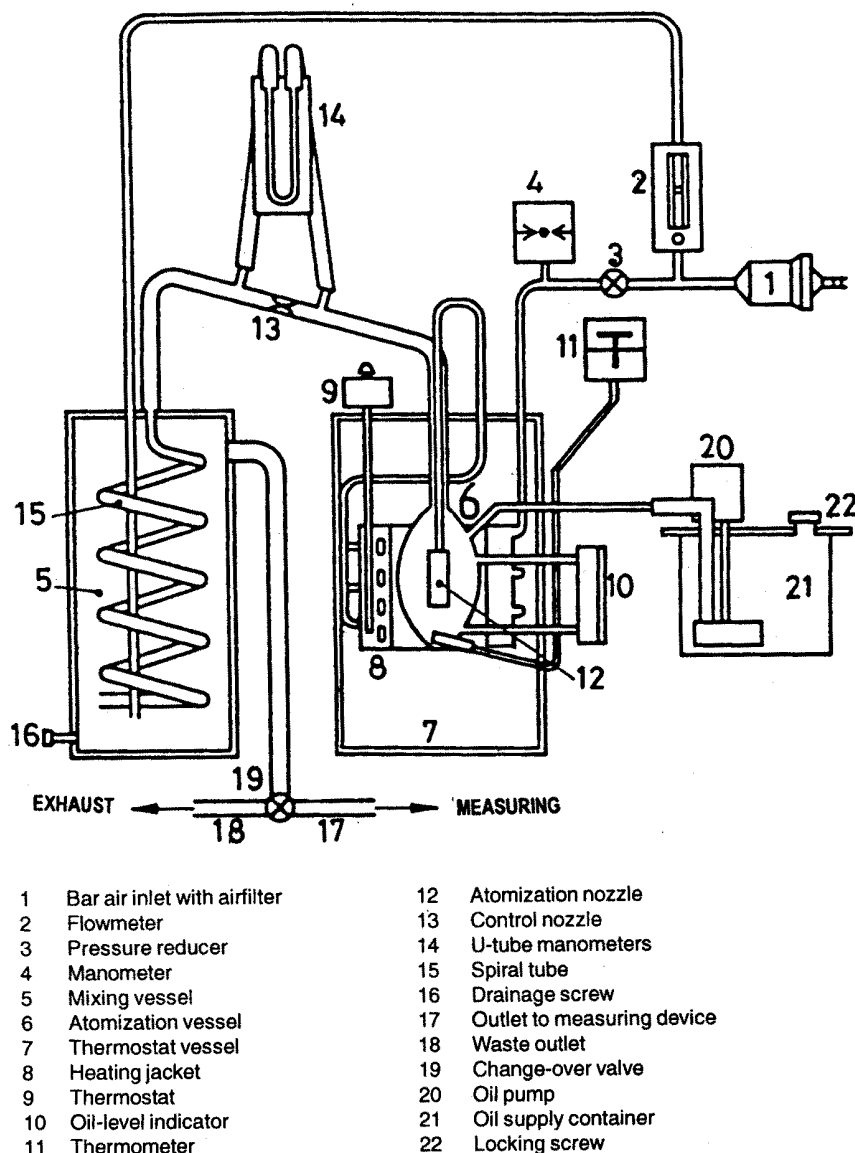


FIG. 10 GENERATOR FOR PARAFFIN OIL AEROSOL

#### A-4.3.3 Test Procedure

The test aerosol is fed into the test chamber, where the filter under test is fixed. A flow rate of 95 l/min is sucked through the filter by means of a suitable pump. The aerosol concentration is measured immediately before and after the filter by the aerosol photometer. The initial penetration shall be measured at  $3 \pm 0.5$  min.

#### A-4.3.4 Calculation of the Penetration

$$P = \frac{I_2 - I_0}{I_1 - I_0} \times 100 \text{ percent}$$

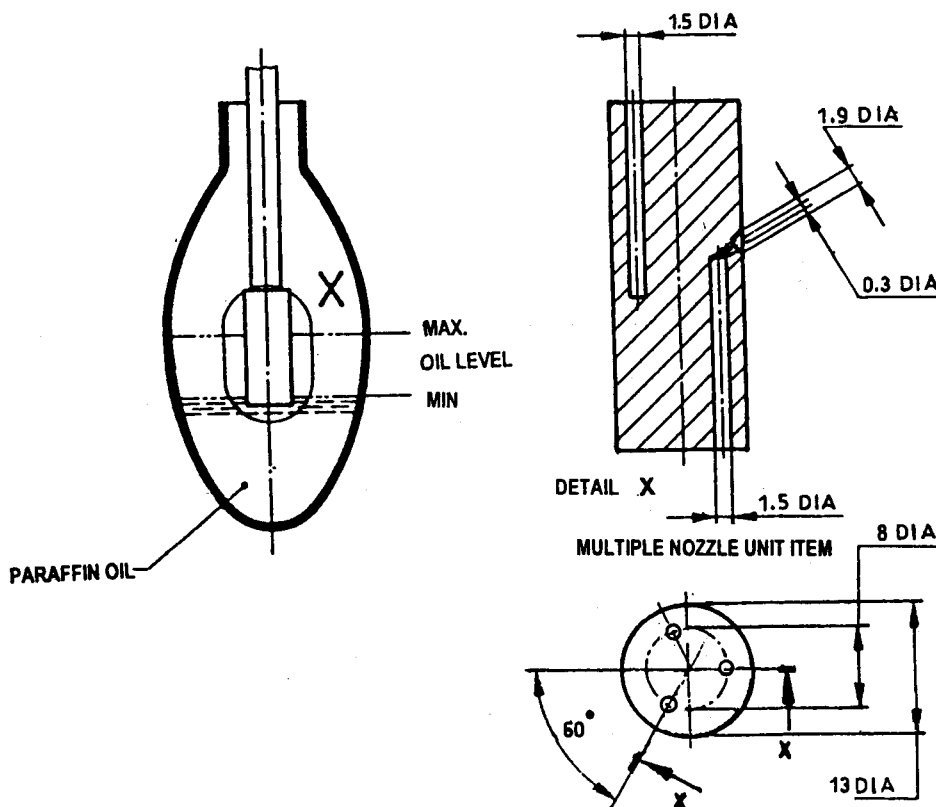
where

- $P$  = Penetration,  
 $I_1$  = Photometer reading before the filter,  
 $I_2$  = Photometer reading after the filter, and  
 $I_0$  = Photometer zero reading for clean air.

#### A-5 FLAMMABILITY

**A-5.1** A total of 4 filtering half masks shall be tested; 2 as received and 2 after temperature conditioning in accordance with A-1.2.

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All dimensions in millimetres.

FIG. 11 ATOMIZER

**A-5.2** The filtering half mask shall be tested for flammability as per Annex E of IS 14166, but using only one burner and with the following variations:

- For the test the filtering half mask shall be put on the metallic dummy head which shall be motorized such that it can describe a horizontal circle with variable control speed. The head shall pass over a Bunsen propane burner whose position can be adjusted by means of suitable gauges, the height of the flame shall be set to 40 mm and the distance between the top of the burner and the lowest part of the filtering half mask passing the filtering half mask through the flame shall be set to 20 mm.
- The head shall be set in motion and the effect of passing the filtering half mask through the flame shall be noted.
- The temperature of the flame at a height of 20 mm above the burner tip shall be  $800 \pm 50^\circ\text{C}$ . The temperature shall be checked with a suitable measuring instrument.

- For the test, the filtering half mask shall be passed once through the flame at  $6 \pm 0.5$  cm/s. When components such as valve(s), etc are arranged on other parts of the filtering half mask, the test shall be repeated with other samples of the filtering half mask in the appropriate position.

## A-6 CARBON DIOXIDE CONTENT OF THE INHALATION AIR

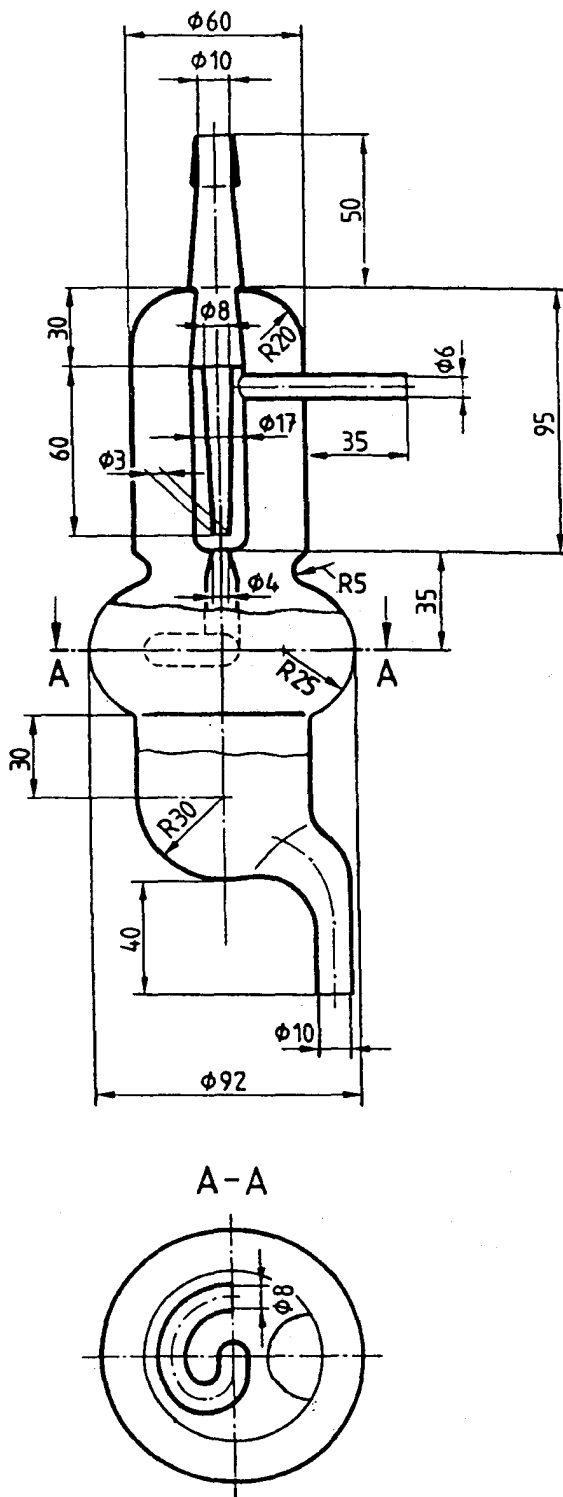
**A-6.1** A total of 3 filtering half masks shall be tested, all 3 as received.

**A-6.2** The apparatus consists essentially of a breathing machine with solenoid valves controlled by the breathing machine, a connector, a  $\text{CO}_2$  flow meter, a  $\text{CO}_2$  analyzer and a humidifier.

The apparatus subjects the filtering half mask to a respiration cycle by the breathing machine.

For this test, the filtering half mask shall be fitted securely in a leak-tight manner but without deformation to the dummy head (type Sheffield) (see Fig.15).

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All dimensions in millimetres.

FIG. 12 BLOWERS ACTUATED BY AIR POWERS

Air shall be supplied to it from a breathing machine adjusted to 25 cycles/min and 2.0 l/stroke and the exhaled air shall have a carbon dioxide content of 5 percent by volume.

A typical test arrangement is shown in Fig. 16.

If the design of the test equipment causes a CO<sub>2</sub> build up, a CO<sub>2</sub> absorber shall be used in the inhalation branch between solenoid valve and breathing machine.

The CO<sub>2</sub> is fed into the breathing machine via a control valve, a flow meter, a compensating bag and two non-return valves.

Immediately before the solenoid valve a small quantity of exhaled air is preferably continuously withdrawn through a sampling line and then fed into the exhaled air via a CO<sub>2</sub> analyzer.

To measure the CO<sub>2</sub> content of the inhaled air, 5 percent of the stroke volume of the inhalation phase of the breathing machine is drawn off at the marked place by an auxiliary lung and fed to a CO<sub>2</sub> analyzer. The total dead space of the gas path (excluding the breathing machine) of the test installation should not exceed 2 000 ml.

The CO<sub>2</sub> content of the inhaled air shall be measured and recorded continuously.

#### A-6.3 Test Conditions

Ambient

atmosphere : laboratory ambient

Exhalation air : 37 ± 2°C, ≥ 95 percent  
relative humidity

Air flow from

the front : 0.5 m/s

For test

arrangement : see Fig. 17

This test shall be performed until a constant carbon dioxide content in the inhalation air is achieved.

#### A-7 STRENGTH OF ATTACHMENT OF EXHALATION VALVE HOUSING

A-7.1 A total of three filtering half masks shall be tested. One as received, one temperature conditioned in accordance with A-1.2 and one after the test described for mechanical strength in A-10.

A-7.2 The filtering half mask shall be mounted securely to a fixture as shown in Fig. 18. An axial tensile force of 10 N shall be applied to the valve (housing) for 10 s. The results shall be noted.

#### A-8 BREATHING RESISTANCE

##### A-8.1 Test Samples and Fixture

A total of 9 filtering half masks shall be tested; 3 as

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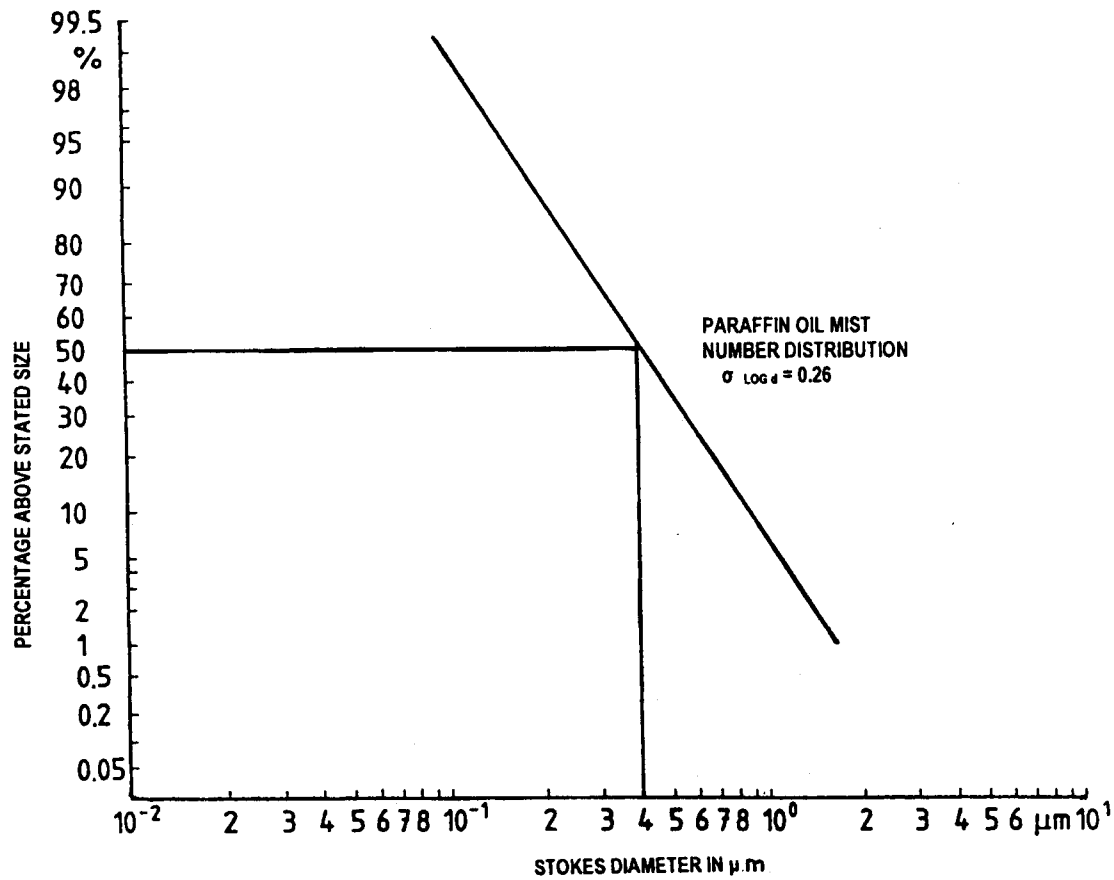


FIG. 13 PARTICLE SIZE DISTRIBUTION OF PARAFFIN OIL MIST

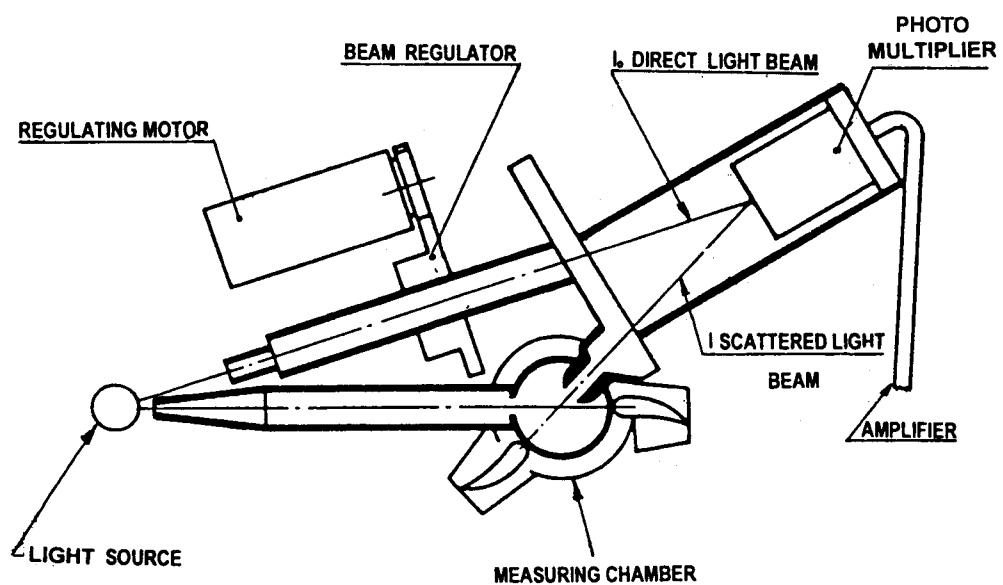
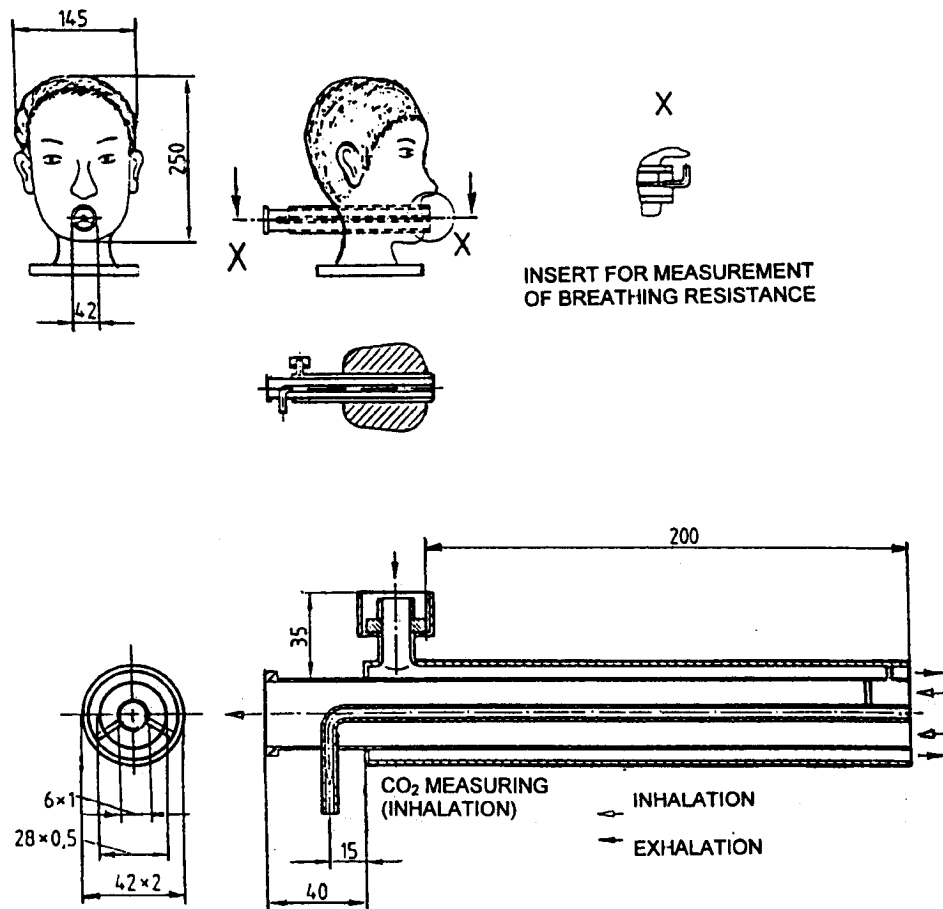


FIG. 14 SCHEME FOR THE AEROSOL PHOTOMETER

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All dimensions in millimetres.

FIG. 15 DUMMY HEAD (SHEFFIELD HEAD) FOR CARBON DIOXIDE CONTENT TEST OF THE INHALATION AIR (DEAD SPACE) FOR A FILTERING HALF MASK

received, 3 after temperature conditioning in accordance with A-1.2, and 3 after the simulated wearing treatment as described in A-1.1.

The filtering half mask shall be fitted securely in a leak tight manner but without deformation to the dummy head (type Sheffield).

#### A-8.2 Exhalation Resistance

The filtering half mask shall be sealed on the dummy head (type Sheffield). The exhalation resistance shall be measured at the opening for the mouth of the dummy head using the adaptor shown in Fig. 15 and a breathing machine adjusted to 25 cycles/min and a 2.0 l/stroke or a continuous flow of 160 l/min. A suitable pressure transducer shall be used.

#### A-8.3 Inhalation Resistance

The inhalation resistance shall be tested at 30 and 95 l/min continuous flow.

#### A-8.4 Correction

The resistance value shall be corrected to 27°C and 1 bar absolute.

#### A-9 CLOGGING

The test aerosol shall be dolomite or coal. A total of 3 filtering half masks shall be tested. One as received and two after temperature conditioning in accordance with A-1.1.

##### A-9.1 Clogging Test Using Coal

##### A-9.1.1 Principle

The test consists of drawing dry, dust laden air through a filtering half mask on the inhalation cycle of a sinusoidal breathing machine and blowing moist, dust free air through the filtering half mask during the exhalation cycle and determining the pressure drop across the filtering half mask following the collection of 1.5 g of dust. The test shall be conducted with the inhaled air at an ambient temperature of  $27 \pm 2^\circ\text{C}$  and

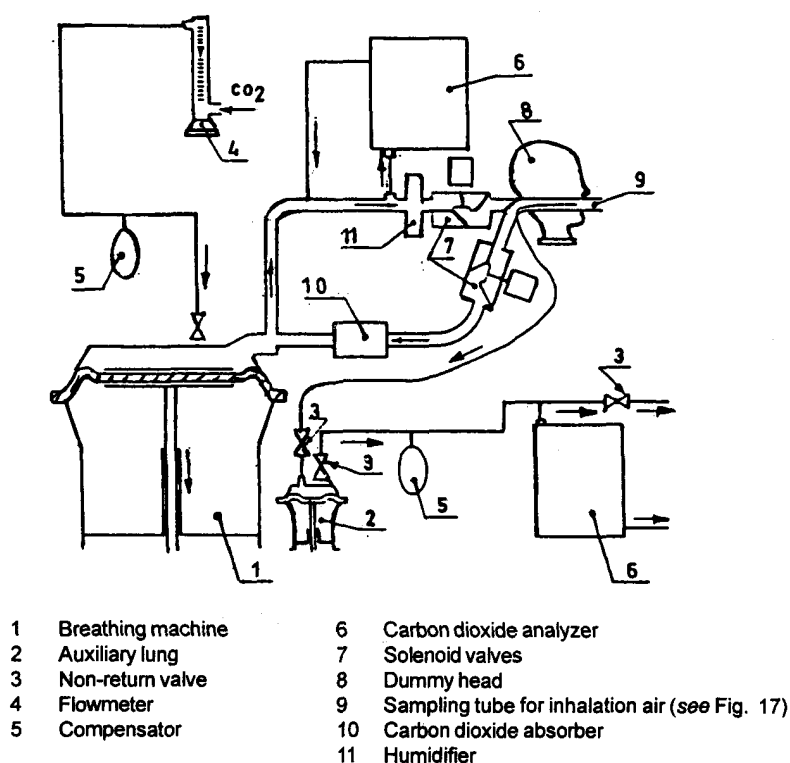


FIG. 16 SCHEME OF TYPICAL TEST RIG FOR CARBON DIOXIDE CONTENT OF THE INHALATION AIR

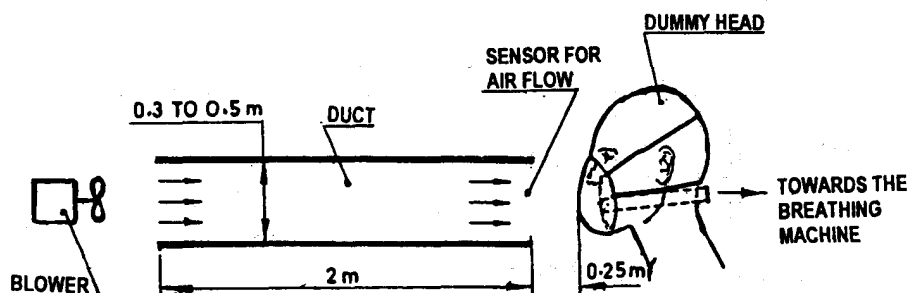


FIG. 17 SCHEME OF TEST ARRANGEMENT AND AIR FLOW FOR CARBON DIOXIDE TEST

at  $65 \pm 5$  percent relative humidity and the exhaled air at  $37 \pm 2^\circ\text{C}$  and 95 percent relative humidity minimum.

#### A-9.1.2 Test Equipment

A scheme of a typical apparatus is given in Fig. 19. The double acting breathing machine *B* (2 l/stroke, 15 cycles/min) and the critical orifice *O* in conjunction maintain the flow rate through mixing chamber *C* and nozzle *A* essentially constant at 95 l/min and 9.5 l/min respectively throughout the sinusoidal breathing cycle, so ensuring dust conditions within the chamber *C*.

For measuring the pressure drop and setting up flow through nozzle *A*, valve *X* is set at CALIBRATE; during these operations, the breathing machine is switched off. For testing the valve is set at TEST. For determining the quantity of dust which shall be introduced into the hopper *H* in order that 1.5 g be collected on the filtering half mask, valve *Y* is set at CALIBRATE; at this setting the flow rate during exhalation is zero.

#### A-9.1.3 Test Conditions

The dust used for the clogging test shall be coal dust

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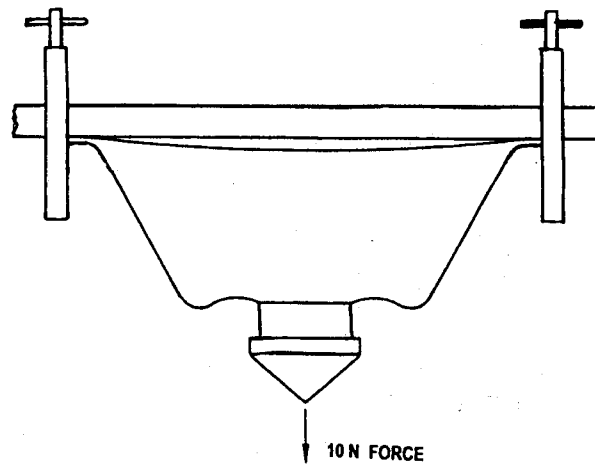


FIG. 18 FIXTURE FOR FACE BLANK

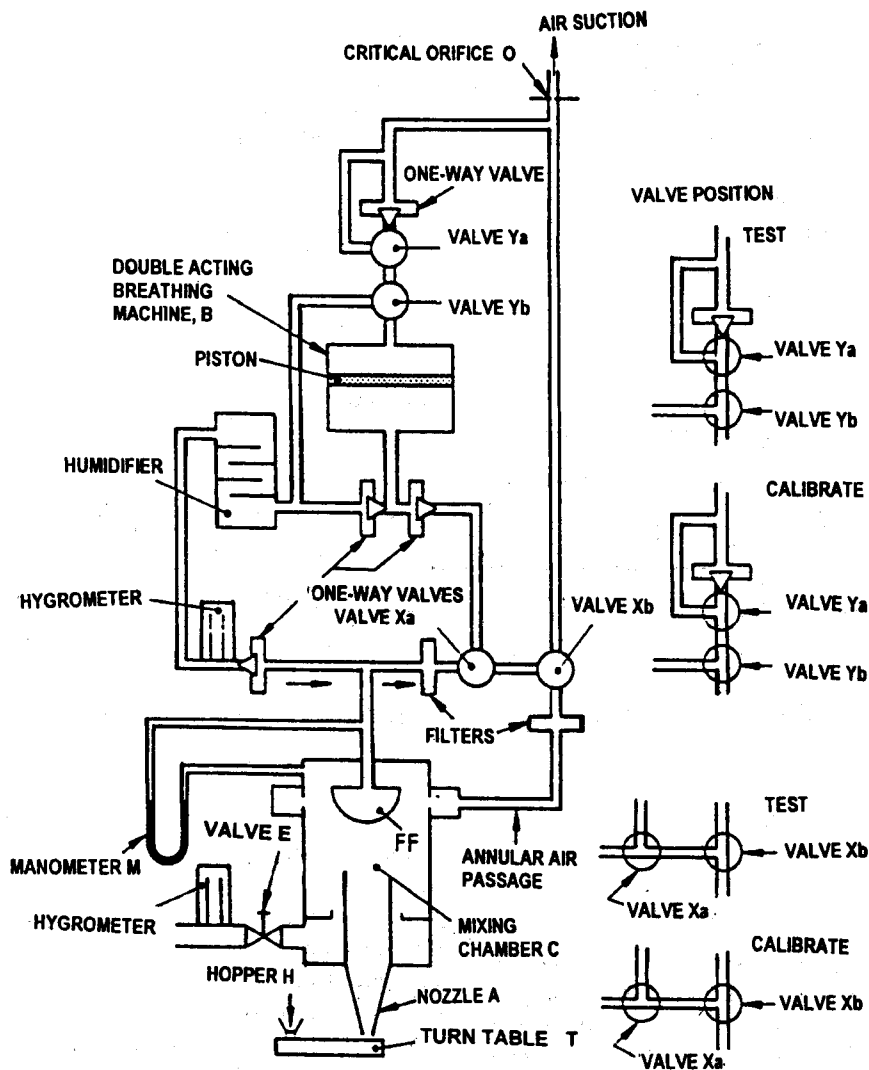


FIG. 19 TYPICAL SINUSOIDAL DUST CLOGGING TEST APPARATUS FOR VALVELESS FILTERING HALF MASKS



ground by ball mill and graded by sieving, the portion which passes a 240 mesh (60 micrometre aperture) sieve being used for the test. The size distribution of coal dust is given in Table 5.

**Table 5 Size Distribution of Coal Dust**  
(Clause A-9.1.3)

| Sl No. | Coulter Counter                               |                                                                                 | Sedimentation Analysis       |                            |
|--------|-----------------------------------------------|---------------------------------------------------------------------------------|------------------------------|----------------------------|
|        | Size<br>(Equivalent<br>spherical<br>diameter) | Percent<br>number<br>particles<br>oversize<br>(No. > 2.5 $\mu\text{m}$<br>=100) | Size<br>(Stokes<br>diameter) | Percent weight<br>oversize |
| (1)    | $\mu\text{m}$<br>(2)                          | (3)                                                                             | $\mu\text{m}$<br>(4)         | (5)                        |
| i)     | 2.5                                           | 100                                                                             | 2.5                          | 89.0                       |
| ii)    | 3.0                                           | 65                                                                              | 3.0                          | 87.5                       |
| iii)   | 5.0                                           | 27                                                                              | 5.0                          | 80.5                       |
| iv)    | 7.0                                           | 14.6                                                                            | 7.0                          | 77.0                       |
| v)     | 10.0                                          | 7.7                                                                             | 10.0                         | 65.0                       |
| vi)    | 15.0                                          | 3.5                                                                             | 15.0                         | 52.5                       |
| vii)   | 20.0                                          | 1.9                                                                             | 20.0                         | 36.0                       |
| viii)  | 25.0                                          | 1.1                                                                             | 25.0                         | 31.2                       |
| ix)    | 30.0                                          | 0.6                                                                             | 30.0                         | 23.2                       |
| x)     | 35.0                                          | 0.3                                                                             | 35.0                         | 15.7                       |
| xi)    | 40.0                                          | 0.2                                                                             | 40.0                         | 9.3                        |

#### A-9.1.4 Test Procedure

The test apparatus shall be set up connecting a weighed filtering half mask into the air circuit at the top of chamber *C*. Valve *X* shall be set to CALIBRATE and air shall be drawn through the filtering half mask at 95 l/min; valve *E* shall be adjusted until the flow through the nozzle *A* is 9.5 l/min. Valve *X* shall then be set to TEST and valve *Y* to CALIBRATE, a known weight of dust shall be fed into the hopper *H* and dispersed into the chamber. After dispersal, the filtering half mask shall be re-weighed. A number of filtering half masks shall be exposed until the weight of dust which shall be introduced into the hopper to give a 1.5 g collection is determined.

Connect the filtering half mask into the chamber. Valve *Y* shall be set to TEST, the previously determined weight of dust shall be introduced into the hopper *H* and after allowing the filtering half mask to reach equilibrium the dust shall be dispersed into chamber *C*. After dispersal valve *X* shall be set to CALIBRATE and the breathing machine turned off. The pressure drop across the filtering half mask at 95 l/min shall be read off manometer *M*.

Repeat the test on the other two samples and report the 3 recorded pressure drops. After completion of the clogging test, the filtering half masks shall be allowed to stabilize in the laboratory for 24 h at the end of which the filtration efficiency of each filtering half mask shall be determined.

#### A-9.2 Clogging Test Using Dolomite Dust

##### A-9.2.1 Principle

The test consists of subjecting the filtering half mask to a sinusoidal breathing simulation. The sample is surrounded by a known concentration of dolomite dust in air. Following the exposure, the breathing resistance and the filter penetration of the sample filtering half mask are measured.

##### A-9.2.2 Test Equipment

A scheme of a typical apparatus is given in Fig. 20. The working area of the test chamber has a suggested square section of 650 mm  $\times$  650 mm. The breathing machine has a displacement of 2 l/stroke. The exhaled air shall pass a humidifier in the exhaled air circuit, such that the exhale air temperature, measured at the position of the sample filtering half mask is  $37 \pm 2^\circ\text{C}$  and 95 percent relative humidity minimum.

##### A-9.2.3 Test Conditions

- DUST:DRB 4/15 dolomite. The size distribution of dolomite dust is given in Table 6.
- The particle size distribution of the air borne dust at the working area of the dust chamber is given in Fig. 21. This characteristic is an essential parameter, which shall be verified especially if the geometry of the test chamber is somewhat different from the model described in this standard.
- Continuous flow through the dust chamber: 60 m<sup>3</sup>/h, linear velocity 4 cm/s.
- Sinusoidal flow through the filtering half mask is delivered by a breathing machine adjusted to 15 cycles/min and 2.0 l/stroke; the exhaled air shall be saturated in humidity.
- Concentration of the dust :  $400 \pm 100 \text{ mg/m}^3$
- Temperature of the air :  $27 \pm 2^\circ\text{C}$ .
- Relative humidity of the air :  $45 \pm 15$  percent
- Testing time : Until the product of measured dust concentration and exposure time is 833 mg h/m<sup>3</sup> or until:
  - for valved filtering half masks, the peak inhalation resistance (corresponding to a continuous flow of 95 l/min) has reached 4 mbar for class FFP1 or 5 mbar for class FFP2 or until the peak exhalation resistance has reached 1.8 mbar (corresponding to 3 mbar at a continuous flow of 160 l/min); and
  - for valveless filtering half masks, the

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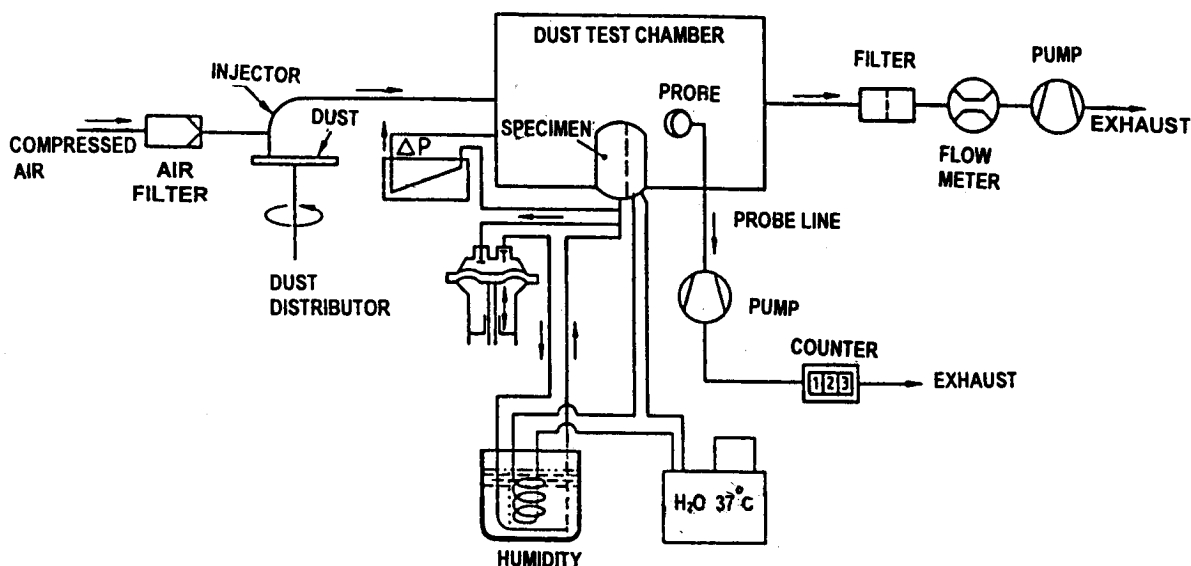


FIG. 20 TYPICAL SINUSOIDAL DOLOMITE DUST CLOGGING TEST APPARATUS

peak inhalation resistance or the peak exhalation resistance has reached 3 mbar for class FFP1 or 4 mbar for class FFP2.

NOTE — 833 mg h/m<sup>3</sup> corresponds to inhaling a total volume of air laden with 1.5 g of dust. This is represented, for example, by a dust concentration of 400 mg/m<sup>3</sup> and an exposure time is 125 min. Because of the dust losses on exhalation the cumulative weight of dust collected on the filtering half mask will probably be less than 1.5 g. For this reason, there is no purpose in weighing the sample filtering half mask.

#### A-9.2.4 Test Procedure

Dust from the distributor shall be conveyed to the dust chamber where it shall be dispersed into the air stream of 60 m<sup>3</sup>/h.

The sample filtering half mask shall be fitted in a leak tight manner to a dummy head or a suitable filter holder located in the dust chamber. The breathing machine and humidifier shall be connected to the sample and shall be operated for the specified testing time.

The concentration of dust in the test chamber may be measured by drawing air at 2 l/min through a sampling probe equipped with a pre-weighed, high efficiency filter (open face, diameter 37 mm) located near the test sample as shown in Fig. 22.

The dust concentration shall be calculated from the weight of dust collected, the flow rate through the filter and the time of collection. Other suitable means may be used.

#### A-9.2.5 Assessment of Clogging

Following the exposure the breathing resistance of the

filtering half mask shall be measured using clean air. The filter penetration shall then be measured in accordance with A-10.

Table 6 Size Distribution of Dolomite Dust  
(Clause A-9.2.3)

| Sl No. | Coulter Counter                               |                                                                      | Sedimentation Analysis       |                            |
|--------|-----------------------------------------------|----------------------------------------------------------------------|------------------------------|----------------------------|
|        | Size<br>(Equivalent<br>spherical<br>diameter) | Percent<br>number<br>particles<br>oversize<br>(No. > 2.5 µm<br>=100) | Size<br>(Stokes<br>diameter) | Percent weight<br>oversize |
|        | µm                                            |                                                                      | µm                           |                            |
| (1)    | (2)                                           | (3)                                                                  | (4)                          | (5)                        |
| i)     | 0.7                                           | 100                                                                  | 1                            | 99.5                       |
| ii)    | 1                                             | 80                                                                   | 2                            | 97.5                       |
| iii)   | 2                                             | 30                                                                   | 3                            | 95                         |
| iv)    | 3                                             | 17                                                                   | 5                            | 85                         |
| v)     | 5                                             | 7                                                                    | 8                            | 70                         |
|        |                                               |                                                                      | 10                           | 50                         |
| vi)    | 9                                             | 2                                                                    | 12                           | 26                         |
|        |                                               |                                                                      | 14                           | 10                         |
| vii)   | 12                                            | 1                                                                    | 18                           | 1                          |

### A-10 MECHANICAL STRENGTH

#### A-10.1 Apparatus

The apparatus, as shown schematically in Fig. 23, consists of a steel case (K) which is fixed on a vertically moving piston (S), capable of being lifted up 20 mm by a rotating cam (N) and dropping down onto a steel plate (P) due to its own mass as the cam rotates. The mass of the steel case shall be more than 10 kg.

### A-10.2 Procedure

**A-10.2.1** The filters shall be treated as received, removed from their packaging but still sealed. The filter shall be placed on their sides in the case (K) so that they do not touch each other during the test, allowing 6 mm horizontal movement and free vertical movement.

**A-10.2.2** After the test any loose material that may have been released from the filter shall be removed

prior to the performance testing. The test rig shall be operated at the rate of approximately 100 rotations/min for approximately 20 min and a total of 2 000 rotations.

### A-11 VISUAL INSPECTION

The visual inspection is carried out where appropriate by the test station prior to laboratory or practical performance tests. The visual inspection includes marking and instructions for use.

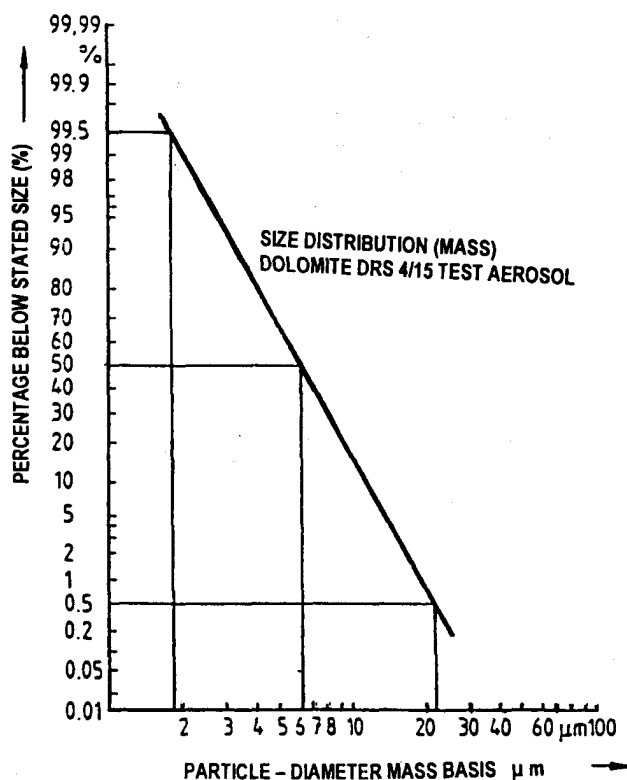
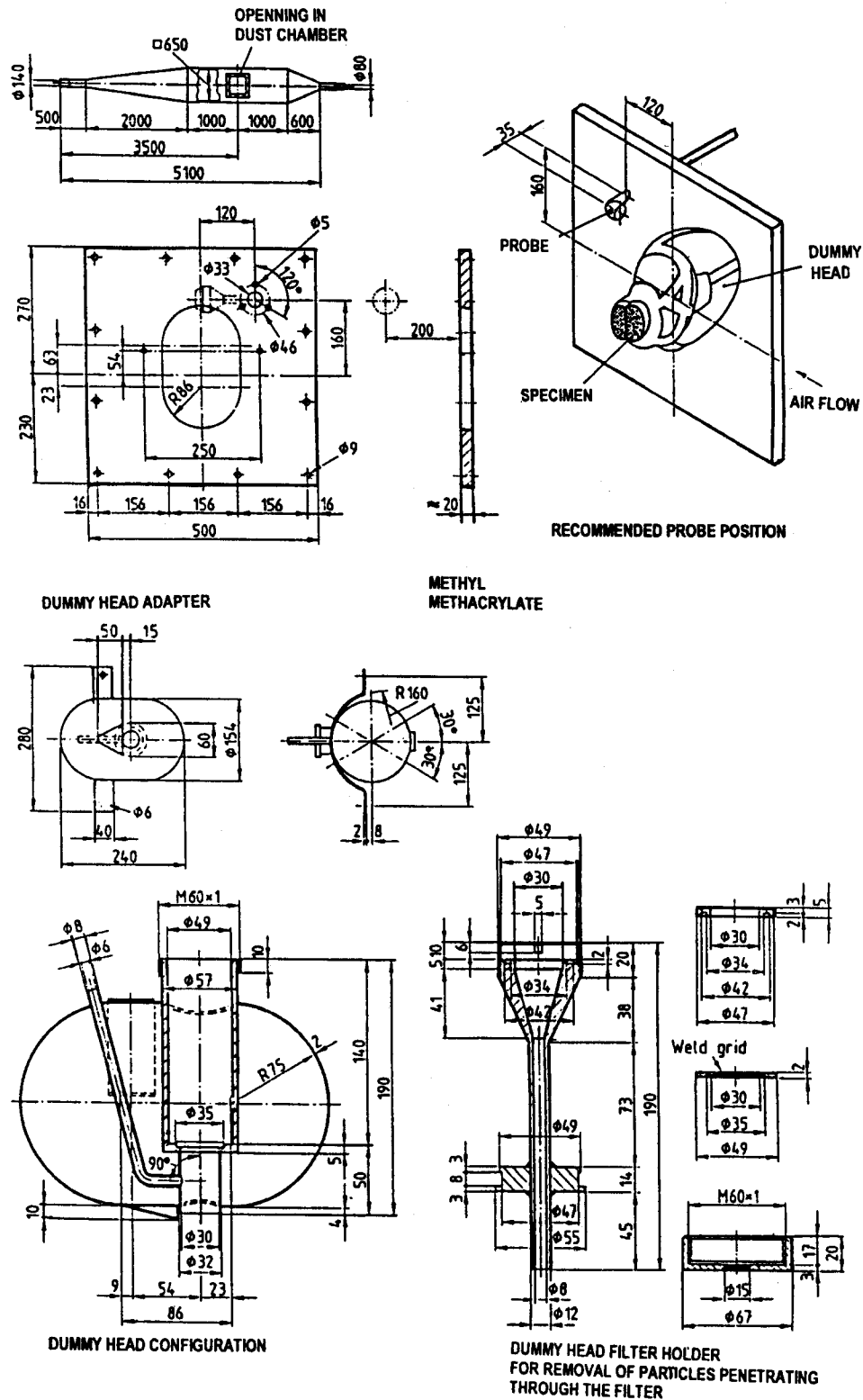


FIG. 21 PARTICLE SIZE DISTRIBUTION OF DOLOMITE DUST IN THE TEST CHAMBER

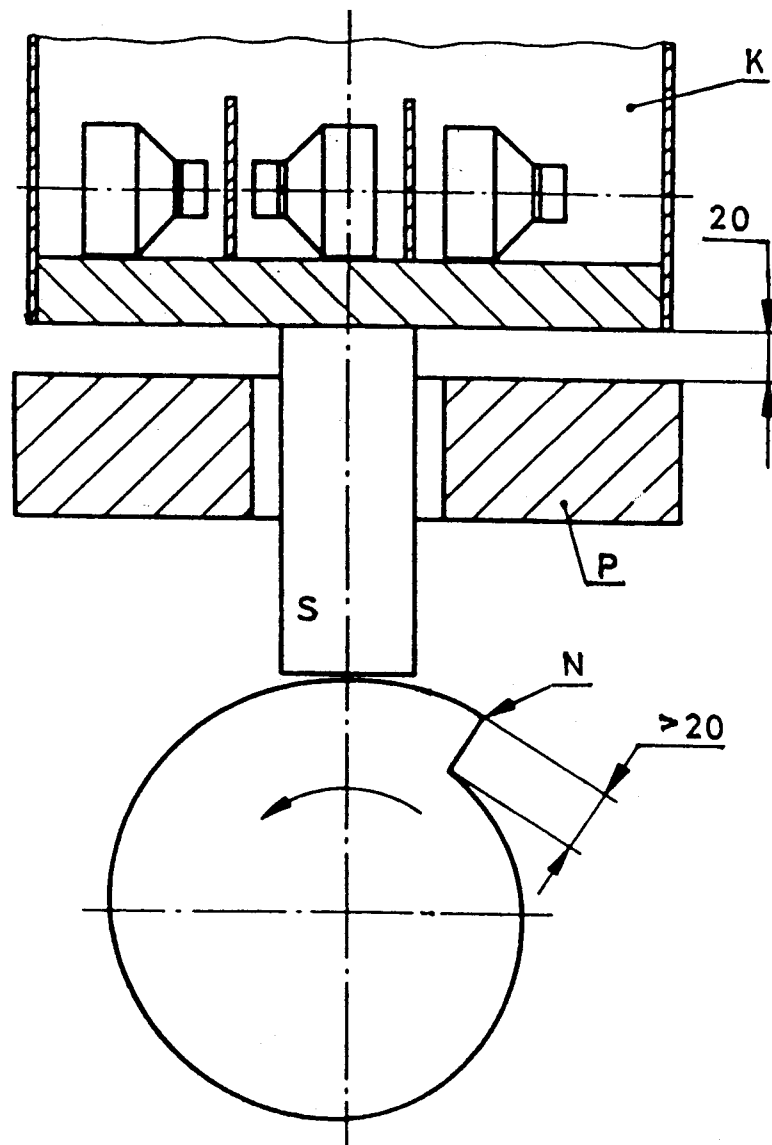
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All dimensions in millimetres.

FIG. 22 DETAILS OF TYPICAL DOLOMITE DUST CLOGGING TEST APPARATUS

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All dimensions in millimetres.

FIG. 23 TEST EQUIPMENT FOR TEST OF MECHANICAL STRENGTH

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## ANNEX B

### (Foreword)

#### COMMITTEE COMPOSITION

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| <i>Organization</i>                                              | <i>Representative(s)</i>                                                                             |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
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